

GSFC NEON Series-1 CMO

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Near Earth Orbit Network (NEON) Series-1 Project, Code 473

**Near Earth Orbit Network
(NEON) Series-1 Project
Sounder for Microwave-Based Applications
Statement of Work**

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Sounder for Microwave-Based Applications
Statement of Work
Review/Signature/Approval Page**

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

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Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0405 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0003 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 17, 2025	This version incorporates NS1-CCR-0011 which was approved by the NS1 SMBA CCB on the effective date shown. Editorial change all instances of climate to weather.

Table of TBDs/TBRs/TBSs

Item No.	Location	Summary	Individual/ Organization	Due Date
1	VI.	Applicable Documents - Data Interface Requirement Document (IRD) (TBS)	Instrument Provider	
2	VI.	Applicable Documents - SMBA Contamination Control Plan (CCP) (TBS)	Instrument Provider	
3	VI.	Applicable Documents - SMBA FVTS Requirements Specification (TBS)	Instrument Provider	
4	VI.	Applicable Documents - SMBA FVTS Interface Requirements Control Document (IRCD) (TBS)	Instrument Provider	
5	VI.	Applicable Documents - Mission Assurance Compliance Matrix for the Sounder for Microwave-Based Applications (SMBA) (TBS)	Instrument Provider	
6	VII.	Reference Documents - SMBA Flight Software and Table Update Delivery Process (TBS)	Instrument Provider	
7	VII.	Reference Documents - SMBA Continuous Risk Management Plan (TBS)	Instrument Provider	
8	VII.	Reference Documents - Drill Template Design Guidelines (TBS)	Instrument Provider	
9	VII.	Reference Documents - SMBA Flight Spare Philosophy (TBS)	Instrument Provider	
10	1.1.1	Government Oversight - The Contractor shall ensure that all software documentation, source code and software design models required for a NASA Software Independent Verification and Validation (IV&V) effort are made available to NASA IV&V personnel as required by the Government (TBR).	Instrument Provider	
11	1.1.2	Government Resident Office Support - The Contractor shall provide facilities to support nominally TBS Government on-site representatives at the instrument development/build site, including private securable office space, furniture, file/storage area, telephones, and printer or printer access, as well as access to a shared conference room from Contract award through satellite on-orbit acceptance (nominally Launch + 6 months).	Instrument Provider	

12	1.1.3	Quality Management System - The developer shall implement a Quality Management System as required by the SMBA IMAR document, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS. This document can be found in Attachment TBS of the Contract.	Instrument Provider	
13	1.1.4.1 d)	Incorporation of Lower Tier Supplier Information - Problem Reports per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS, typically summarized at the PMR as required.	Instrument Provider	
14	1.3	Risk Management - The Contractor shall establish and maintain a comprehensive risk management program that meets the requirements of Contractor Risk Management Plans and the NEON Program Continuous Risk Management Plan (TBS).	Instrument Provider	
15	Table 1-1.	Top-Level Instrument Reviews - CDRL References TBS	Instrument Provider	
16	1.5.1.13	Technical Interchange Meetings and Other Reviews and Meetings - Per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS, the Contractor shall invite the Government to participate in the Contractor's Failure Review Board (FRB) and Material Review Board (MRB).	Instrument Provider	
17	1.5.1.14	Subcontract and Subsystem Reviews - The Contractor shall submit Subsystem/Subcontract Review Data Packages in accordance with CDRL (TBD).	Instrument Provider	
18	3.1	Deviations and Waivers - Requests for deviations or waivers to the Government's requirements shall be submitted to the Government's Contracting Officer for Government approval per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (NS1-SMA-REQ-	Instrument Provider	

		0002) within Contract TBS, Clause J.1, Attachment N (TBR), and CDRL SE-1.		
19	4.2.11.1	Test and Verification - The Contractor shall plan and implement the testing of instrument-spacecraft interfaces with the Spacecraft-Instrument Interface Simulator (SIIS) provided by the spacecraft Contractor (TBR). Technical support will be provided by the spacecraft Contractor at the instrument simulator provider facility. This will support early risk reduction of instrument-spacecraft interfaces.	Instrument Provider	
20	4.2.12	NASA Independent Verification and Validation Support - The Contractor shall ensure that all software documentation and code required for NASA's Software IV&V effort is made available to NASA IV&V personnel if directed by the Government (TBR).	Instrument Provider	
21	4.2.13	Software Maintenance - The delivery of a flight software patch or update shall be executed via the processes specified in the Project Flight Software and Table Update Delivery Process (TBS) (i.e., by sending a CD with the patch or uploading to a SharePoint folder).	Instrument Provider	
22	4.2.15	Flight Software Deliverable - The Contractor shall deliver and periodically update a Flight Software Development Environment (CDRL SW-TBD)	Instrument Provider	
23	4.3.3	Ground Support Equipment - RFI generator (TBR).	Instrument Provider	
24	4.3.3	Ground Support Equipment - The Contractor shall provide two (TBR) additional sets of thermal control equipment for exclusive use at the spacecraft integration facility(ies).	Instrument Provider	
25	5	Test Program - Contractor shall take and maintain test data in accordance with the Sensor Performance Specification, the approved test plan (CDRL IT-1) and the requirements. Problems, anomalies, failures, etc., shall be reported and treated in accordance with the IMAR document, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract	Instrument Provider	

		TBS, Clause TBS, Attachment TBS, identified in Section IV for investigation, resolution, and corrective action.		
26	5.6.2	Instrument Calibration - The Contractor shall perform calibration at a minimum of three instrument temperatures representing the SMBA instrument low operational, mid operational, and high operational temperatures (TBS).	Instrument Provider	
27	5.6.2	Instrument Calibration - Intercalibration may be done at GSFC (TBR).	Instrument Provider	
28	6.5	Sensor Database Support - The Contractor shall supply XML (TBR) formatted database inputs as required by the ground system.	Instrument Provider	
29	6.6.5.6 a)	Mission Rehearsals - Three (3) (TBR) instrument-focused rehearsals and dress rehearsals.	Instrument Provider	
30	6.6.5.6 b)	Three (3) (TBR) Support Operations Readiness Exercises as needed to prepare for and/or supplement Mission Rehearsals.	Instrument Provider	

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I. INTRODUCTION

The Sounder for Microwave-Based Applications (SMBA) is a passive microwave instrument that produces global atmospheric temperature and moisture profiles. SMBA provides microwave data to support temperature and humidity sounding data in cloud covered and clear-sky conditions.

The Instrument has a risk classification of tailored Class C as per Goddard Procedural Requirement (GPR) 8705.4A.

This Statement of Work (SOW) covers the work required to develop, fabricate, integrate, test, analyze, verify, deliver, and provide post-launch support for the SMBA instruments required for the Near Earth Orbit Network (NEON) Program.

II. SCOPE

This SOW reflects the Contractor's scope and period of performance as defined in the Contract. Throughout the remainder of the document, the instrument supplier will be referred to as the Contractor. The National Aeronautics and Space Administration (NASA)/Goddard Space Flight Center (GSFC), including its support services Contractors, will be referred to as the Government. This SOW and the Contract Data Requirements List (CDRL) (NS1-SMBA-CDRL-0001) identify the deliverables associated with the SMBA instrument. The Contractor **shall** meet the requirements of all contractual documents. The Contractor **shall** facilitate insight by the GSFC Program Office into the instrument progress and status. Compliance with the requirements stated herein **shall** be the responsibility of the Government and the Contractor.

This SOW defines the minimum effort required of the Contractor that **shall** include the business and technical management including necessary related Government reviews and discussions, design, engineering analyses, data reduction, data presentations, technical reviews, development, fabrication, integration, test, evaluation, delivery, and support for the SMBA instruments and software in support of the NEON Program. This support **shall** include post-delivery support from delivery of the SMBA through the period of performance of this Contract. This SOW defines the roles, responsibilities, and obligations between the Government and the Contractor. The Contractor **shall** develop this instrument using sound engineering principles founded in industry best practices, stated Government standards and requirements, and agreed-upon Contractor internal standards and policies.

This SOW covers all SMBA instrument builds. Any requirement that is only applicable to the first flight unit or any other unit is specifically called out. Where no clarification regarding applicability is provided, the requirements are applicable to all flight units.

III. NEON PROGRAM AND NEON SERIES-1 PROJECT OVERVIEW

The NEON Program, a part of the National Environmental Satellite, Data, and Information Service Office of Low Earth Orbit (LEO) Observations, provides a framework to provide global measurements of Earth systems. These environmental measurements will be able to support a wide variety of atmospheric, terrestrial, marine, and polar observations. The data includes the numerical weather prediction models, fire and flood models, atmospheric chemistry

observations, and multiple land imagery products that have been a crucial piece of the National Oceanic and Atmospheric Administration (NOAA) strategic goal to create a "Weather Ready Nation" and which provide essential information to NOAA's broader environmental stewardship mission. Following the recommendations from the 2017 NOAA Satellite Observation System Architecture (NSOSA) study, the NEON Program intends to use a combination of small and medium satellite platforms in a disaggregated architecture. This approach will provide an efficient means of filling gaps quickly and taking advantage of emerging remote sensing technologies.

To execute a disaggregated LEO architecture and other NSOSA recommendations, the NEON Program plans to:

- Use industry's significant investment of funding, expertise, and innovation in space and space systems technology.
- Deploy NOAA observation system assets where and when they are most needed, enabled by shorter development timelines and more frequent launches.
- Leverage smaller instruments, satellites, and evolving small launch vehicles; and
- Achieve greater agility to incorporate continuous advancement, using new business models and partners.

The NEON Program will accomplish these goals by continuing to leverage the multi-decadal partnership between NOAA and NASA.

The NEON Series-1 Project is an initial operational project for the NEON Program featuring the development, integration, test, launch, and science operations of the SMBA instrument – a successor to the Advanced Technology Microwave Sounder (ATMS) instrument, versions of which are on each of the Joint Polar Satellite System (JPSS) satellites. It will allow the NEON Program to provide continuity with the existing JPSS-era microwave sounder measurements as well as leverage newer technological advances to advance the collection of hyperspectral microwave sounding data that is crucial to numerical weather prediction models.

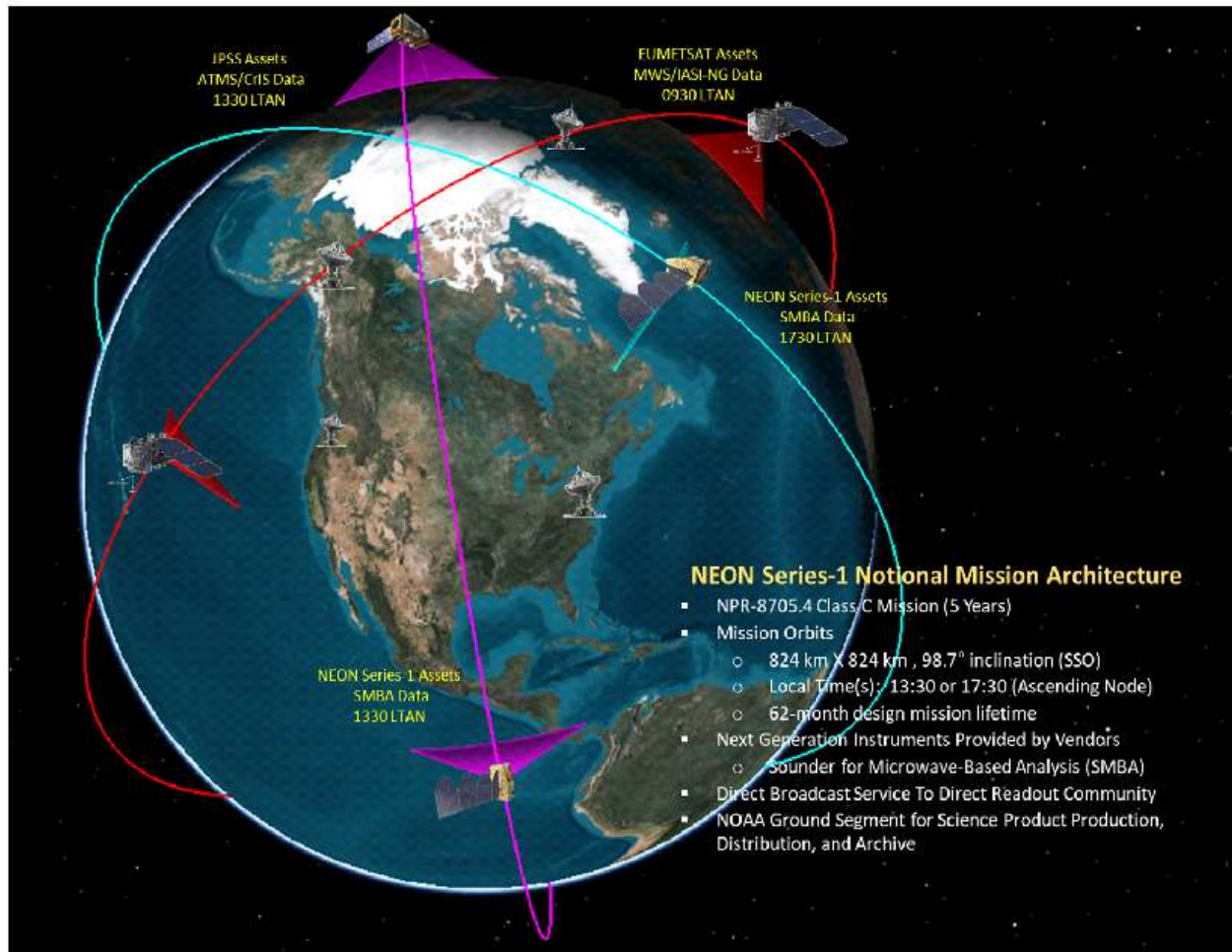


Figure III-1. Notional NEON Series-1 Project Architecture

Utilizing elements and best practices of the commercial space industry to meet federal program level requirements, the NEON Series-1 Project will demonstrate the NEON Program can provide operational hyperspectral microwave sounding observation capabilities with a small satellite that can be flexibly deployed to a variety of operational orbits. This first operational project, consisting of four individual flight missions to two distinct sun synchronous orbits (local time of ascending nodes (LTAN) of 1330 and 1730), will demonstrate the NEON disaggregated mission portfolio that provides the program with the flexibility of placing hyperspectral microwave sounders in either operational orbit (1330 or 1730 LTAN), at any desired launch cadence as the program needs to support hyperspectral microwave sounder needs.

The NEON Series-1 Project will also leverage mission assurance practices relative to established commercial New Space missions and establish a referenceable knowledge base of rapid procurement practices that will benefit the future environmental observation missions planned in the high launch tempo NEON Program.

The NEON Series-1 Project is Class C as per GPR 8705.4, which applies the appropriate NASA management controls, systems engineering processes, mission assurance requirements and risk management processes. Whereas reliability requirements are not levied in this document, select

areas of redundancy should be considered that may offer reasonable improvement to mission success.

IV. THE SOUNDER FOR MICROWAVE-BASED APPLICATIONS

SMBA is the principal instrument aboard the NEON Series-1 satellite. The instrument (hereafter referred to as the “SMBA instrument” or simply the “instrument”), is a cross-track microwave sounder that collects hyperspectral microwave radiometric data that are used to calculate the vertical distribution of temperature (T), moisture (q), and pressure in the Earth’s atmosphere.

The instrument measures atmospheric radiances within the microwave range of approximately 20-300 GHz over a swath width of approximately 2,530 km from an altitude of 824 km.

Hyperspectral capability indicates fine spectral resolution. Fine spectral resolution is beneficial for both science and radio frequency interference (RFI) detection capability. Hyperspectral microwave capability refers to microwave sounding systems with approximately 100 or more channels. The increased number of channels results in a finer spectral resolution than is achieved by traditional microwave sounding techniques over the same bandwidth.

Instrument data is acquired continuously, stored on-board the satellite, and subsequently downlinked to a ground network for capture, preprocessing, and routing to meteorological centers.

The instrument produces Raw Data Records (RDR) in an uncalibrated form. Engineering, instrument telemetry, science data, radiometric, and spectral calibration data is provided to enable the RDRs to be calibrated and geolocated. In addition, diagnostic RDR data may be acquired for testing purposes. The data are sent to the NOAA Ground System where they are used as input to science algorithms that generate Temperature Data Records, Sensor Data Records, and Environmental Data Records.

V. DEFINITIONS

Contractor – The developer of the item of reference. If the term Contractor is used in this document without specific reference to an item (e.g., SMBA, ground support equipment (GSE), etc.), then the term **shall** be interpreted to imply the SMBA Contractor. In all cases, the term Contractor also implies any and all associated suppliers and subcontractors.

Government – Federal Civil Servants and support Contractors (Contractors that represent NASA or are delegates for NASA).

Shall – Compliance by the Contractor is mandatory. Any deviations from these contractually imposed mandatory requirements require the approval of the Contracting Officer.

Should – Preferred good practices are identified by “should” statements. “Should” designates performance, which is requested, but not mandatory. Noncompliance of a “should” statement does not require Government approval but **shall** be addressed at the earliest applicable project milestone.

May – At the discretion of the Contractor or Government.

Will – Designates the intent of the Government.

Applicable documents – The term “applicable documents” denotes Government-prepared and controlled documents and industry standards documents that levy additional requirements on the Contractor. The Contractor is required to comply with the documents, or the tailored version of the documents provided by the Government.

Reference documents – The term “reference documents” denotes documents that amplify or clarify the information presented in this document. The Contractor is not required to meet any requirements listed within these documents.

TBS – The term “to be supplied” (TBS) refers to a document or information that will be developed or provided.

TBR – The term “to be reviewed” (TBR) means that the stated requirements or values will be reviewed for appropriateness jointly by the Contractor and the Government and the information may be changed. The TBR requirement/value remains a requirement until formal process of configuration change review, approval, and Contract modification.

TBP – The term “to be proposed” (TBP) indicates required Contractor fill-ins as part of the solicitation process.

TBD – The term “to be determined” (TBD) identifies a missing requirement or value; indicates further research or analysis is needed to determine the information. All changes to resolve these items will be completed through a formal process of configuration change review, approval, and Contract modification.

Instrument – The SMBA flight hardware, support equipment (i.e., system test equipment (STE)/GSE), software, documentation and/or data necessary to fulfill the requirements of the Contract and this SOW.

Satellite – The term “satellite” refers to the NEON Series-1 spacecraft bus integrated with the flight instrument(s). The term “satellite” is equivalent to the term “satellite” used in other NASA and NEON Program documents.

Class I – A change that affects a configuration item's form, fit, or function. This could include changes to baselined documentation, technical requirements, Contract end items, interfaces, weight, power, data rate allocations, technical risks, or science performance. All proposed Class I changes require Configuration Control Board (CCB) approval.

Class II – A change that does not affect a configuration item's form, fit, or function. This could include changes to documentation only, minor hardware changes, or drawing changes that do not affect a baseline or interface. Class II changes do not require Project CCB review unless they will be written against CM-controlled documents.

VI. APPLICABLE DOCUMENTS

The Applicable Documents listed in this section apply directly to the performance of this SOW to the extent specified and invoked by this SOW. The Contractor **shall** comply with the documents to the extent specified herein. The Contract, Attachment N, provides the current issue revision and issue date as well as any additional Configuration Change Requests (CCR) for the applicable documents.

Document No.	Document Name
NS1-SMBA-CDRL-0001	SMBA Contracts Data Requirements List (CDRL)
NS1-SMA-REQ-0002	SMBA Instruments Mission Assurance Requirements (IMAR)
NS1-SMBA-REQ-0001	SMBA Performance Requirements Document (PRD)
TBS	Data Interface Requirement Document (IRD)
TBS	SMBA Contamination Control Plan (CCP)
TBS	SMBA FVTS Requirements Specification
TBS	SMBA FVTS Interface Requirements Control Document (IRCD)
TBS	Mission Assurance Compliance Matrix for the Sounder for Microwave-Based Applications (SMBA)
GSFC-STD-1001	Criteria for Flight and Flight Support Systems Life Cycle Reviews
GPR7120.4	Risk Management
GPR 8705.4A	Risk Classification Guidelines and Risk-Based SMA Practices for GSFC Payloads and Systems

VII. REFERENCE DOCUMENTS

The documents listed in this section are those documents referenced in the SOW which contain information relating to the work required, but which do not constitute requirements.

Document No.	Document Name
TBS	SMBA Flight Software and Table Update Delivery Process
TBS	SMBA Continuous Risk Management Plan
TBS	Drill Template Design Guidelines
TBS	SMBA Flight Spare Philosophy
NPR 8715.3	NASA Safety Manual
NPR 8000.4	Risk Management Procedural Requirements
GSFC-STD-1000	GOLD Rules
NPR 7123.1	NASA Systems Engineering Processes and Requirements
NASA SP-2016-6105	NASA Systems Engineering Handbook
GSFC-STD-7000	General Environmental Verification Standard (GEVS)
NPR 7150.2	NASA Software Engineering Requirements
GPR 8700.6	Engineering Peer Reviews
GPR 8700.4	Integrated Independent Reviews
NPR 8621.1	NASA Procedures and Guidelines for Mishap Reporting, Investigating, and Recordkeeping

ANSI/EIA748	Industry Guide for Earned Value Management
IEST-STD-CC1246D	Product Cleanliness Levels and Contamination Control Program
470-LEO-PROC-00010	Low Earth Orbit (LEO) Programs Division (LPD) Configuration Management Procedure

VIII. PRECEDENCE

In the case of conflicting requirements, the order of precedence of documents not specifically called out in the Contract is: The Contract takes precedence over the SOW, followed by the CDRL, Instrument Mission Assurance Requirements (IMAR), Performance Requirements Document (PRD), Interface Control Documents (ICD), and the remaining applicable documents. In the event of an inconsistency between any of the reference documents herein and the requirements of this SOW, the SOW **shall** take precedence. Any conflicts discovered by the parties between the requirements in this SOW and any referenced specification herein **shall** be brought to the attention of the other party for resolution.

IX. WORK TO BE PERFORMED

This section, along with the CDRL, GSFC NS1-SMBA-CDRL-0001, describes the specific work to be accomplished by the Contractor. In accordance with the requirements of this document, the Contract, all associated requirements documents, other attachments and applicable documents to this Contract, the Contractor **shall** provide the effort and resources that include, but are not limited to personnel, materials, equipment, and facilities necessary for the successful and on-time design, analysis, development, fabrication, assembly, integration, test, calibration, qualification, and delivery and support of the instrument. The Contractor **shall** also provide the personnel, materials, equipment necessary to support Satellite Integration and Test, support to mission-level activities, training, and post-delivery support of the instrument throughout the performance of this Contract.

The Contractor **shall** deliver to the Government an instrument that is fully tested and calibrated with verified compliance to the requirements of this Contract. The Contractor **shall** support the development of bilateral instrument documents, analyses, and models, which define the details of the instrument-to-spacecraft interface and instrument accommodation and operating information.

1 MANAGEMENT

1.1 Program Management

The Contractor **shall** appoint a dedicated Program Manager (PM) to direct and manage the SMBA instrument development. The Contractor's PM **shall** have responsibility for the overall technical performance, resource management, and schedule management of the contractual effort and all subcontracts. The Contractor's designated PM **shall** report to a level of company management appropriate to ensure prompt resolution of all problems. The PM **shall** maintain close liaison with the designated Contracting Officer's Representative (COR) and Instrument Manager and be responsible for effective and timely communication regarding program performance. The Contractor's PM **shall** ensure that all requirements of this Contract are accomplished successfully and on-time.

1.1.1 Government Oversight

To meet the mission and schedule requirements, the Government intends to establish a collaborative teaming environment with the Contractor to gain adequate insight of designs, Assembly, Integration, and Test activities, and operations. The Contractor's management, tracking, control, and reporting processes **shall** facilitate insight by the GSFC Program Office into the instrument progress and status. The Contractor **shall** provide the Government virtual access, on and off site, to the instrument configuration management, risk management, and planning and scheduling systems. The Contractor **shall** provide virtual access, on and off site, to the instrument anomaly reporting system.

The Government will maintain a SharePoint, or similar, on-line collaborative space where the Contractor may share informal information with the Government.

The Government **shall** have virtual access, from both inside and outside the Contractor's facility, to documents and data, including drawings and the requirements verification database.

The Contractor **shall** open Government attendance to Contractor and subcontractor/lower tier suppliers' internal data, reviews, audits, meetings, and other activities within the scope of the Contract where the Government has determined based on Contractor and/or subcontractor performance and/or risk to the program a need for such oversight. The Contractor **shall** notify the COR and the Government resident office representatives of meetings, reviews, or tests in sufficient time to permit meaningful Government participation.

The Contractor **shall** ensure that all software documentation, source code and software design models required for a NASA Software Independent Verification and Validation (IV&V) effort are made available to NASA IV&V personnel as required by the Government (TBR). The Contractor **shall** allow NASA IV&V review and participation before final product delivery to the Government.

1.1.2 Government Resident Office Support

To facilitate insight, the Government will establish a resident office that will include management, systems engineering, and quality assurance representation. The resident office representatives will attend the Contractor's daily meetings to stay current on activities and issues.

The Government will staff a contributive impactful resident office at the Contractor's facility to enable quick and efficient communication, problem resolution, and requirements clarification as necessary. The Government Resident Office will follow the Contractor's development and observe its processes including assembly, test, and inspections. The intent is to minimize overlaying Government processes and deliverables as much as possible with minimal disruptions to the development schedule. The Contractor **shall** provide facilities to support nominally TBS Government on-site representatives at the instrument development/build site, including private securable office space, furniture, file/storage area, telephones, and printer or printer access, as well as access to a shared conference room from Contract award through satellite on-orbit acceptance (nominally Launch + 6 months). The Contractor **shall** provide within these offices high-speed (broadband) internet access and periodic wireless access, as required, to an internet service provider within the Contractor's facility to allow for access to the GSFC network.

1.1.3 Quality Management System

The developer **shall** implement a Quality Management System as required by the SMBA IMAR document, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS. This document can be found in Attachment TBS of the Contract.

1.1.4 Subcontract Management

The Contractor **shall** actively manage its Lower-Tier Suppliers (LTS) including robust mission and quality assurance focused oversight and technical insight. The Contractor **shall** routinely verify LTS adherence to both Contractor-flowed down and approved subcontractor processes. The Contractor **shall** meet the requirements in the following sections as noted to provide the Government with visibility into LTS activities. The Contractor **shall** maintain a schedule of onsite visits to its key suppliers or those presenting risk to the program including planned and actual surveillance audits and be available for review at each Project Management Review (PMR). The Government may require onsite personnel at LTS facilities in cases of frequent anomalies.

1.1.4.1 Incorporation of Lower Tier Supplier Information

The Contractor **shall** provide to the Government the following LTS information at a minimum:

- a) Earned Value Management System (EVMS) and financial data as required herein.
- b) Schedule and Schedule Reporting, typically via the Integrated Master Schedule (IMS).
- c) Project Status Reviews and/or Reports, typically via the PMR.
- d) Problem Reports per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS, typically summarized at the PMR as required.
- e) Design, Manufacturing, Integration and Test Review schedule, and contact information, minutes, or material discussed.
- f) LTS Risk items, risk mitigation tasks and status of risks **shall** be reported and included in each review if required and/or as requested by the Government.

1.1.4.2 Meetings with Lower Tier Suppliers

The Contractor **shall** invite the Government to all formal/major reviews and acceptance activities at LTSs.

1.1.4.3 Flow Down of Requirements to Lower-Tier Suppliers

The Contractor **shall** flow all applicable requirements and controls stated or referenced herein to its LTSs. The LTSs **shall** supply a compliance matrix delineating their compliance with each of the requirements, provided to the Government upon request.

1.1.4.4 End Item Data Packages from Lower Tier Suppliers

The Contractor **shall** submit End Item Data Packages (EIDP) from LTSs in accordance with CDRL MA-32.

1.2 Financial Management

The Contractor **shall** use and maintain in performance of this Contract an EVMS. The Contractor's system **shall** meet the guidelines and be maintained in accordance with the EVMS requirements described in this Contract, under NASA Federal Acquisition Regulation Supplement Clause 1852.234-2. The EVMS **shall** be supported by the Contractor's management processes and systems and in accordance with the Contractor's EVM Plan. Reference the Contract Performance Report delivered as part of the Integrated Program Management Report (CDRL PM-4).

1.2.1 Performance Measurement Baseline

The Contractor **shall** define the Performance Measurement Baseline (PMB), completing a time-phased integrated plan to unify technical scope, schedule, and budget into an overall plan in accordance with the Contractor's EVM Implementation Plan. The Contractor **shall** define the work scope, establish a schedule for accomplishing the work, and estimate the resources required to accomplish the work. The Government will complete a PMB cost and schedule validation as required.

1.2.2 Integrated Master Schedule

The Contractor **shall** submit an IMS to be used to manage this instrument development effort. (Refer to CDRL PM-4.)

The Contractor **shall** submit and maintain a comprehensive listing of all external Giver-Receiver (G-R) items also referred to as Receive-Deliver items to provide the project a description of each G-R indicated in the IMS. (Refer to CDRL PM-4.)

1.2.3 Work Breakdown Structure

The Contractor **shall** develop and maintain the Work Breakdown Structure (WBS) and WBS dictionary in accordance with CDRL PM-2. The Contractor uses the WBS as the primary framework for Contract planning, budgeting, and reporting of the cost, schedule, and technical performance status to the Government. The WBS structure **shall** be established by the Contractor (based on guidelines provided by the Government) and **shall** be reported to the Government at a minimum of Level 3 decomposition. The Contractor **shall** maintain a WBS at a minimum one level lower than reporting Level 3. Prior notification to the Government of WBS changes at Level 3 or higher **shall** be required before implementation (Reference WBS CDRL PM-2). The WBS through level 3 **shall** be established in conjunction with the NEON Financial Management team prior to submittal of a proposal in accordance with this SOW.

1.2.4 Organization Breakdown Structure

The Contractor **shall** develop and maintain the Organization Breakdown Structure (OBS), outlining hierarchical arrangement of the Contractor's management organization, which human entity is responsible for the work and the associated reporting relationships. The OBS **shall** have the same type of definition as the WBS, accomplished through a charter of functions and responsibilities and is the organizational equivalent to a WBS dictionary.

1.2.5 Change Control

In order to maintain baseline integrity and assure oversight of the baseline change control process, the Contractor **shall** use a formalized change control process for all Baseline Change Requests (BCR) via the CCB process established by the Contractor. The Government **shall** be a member of the BCR CCB and **shall** have approval authority for any BCR over \$250,000 or any schedule BCR that impacts the critical path. BCRs incorporating rolling wave changes and/or new scope do not require Government approval. In the event the Government does not attend a CCB, the Contractor **shall** provide notification, via e-mail submittal, of all BCRs. The Government will respond within three (3) working days, provided that pre-coordination of the BCR has occurred. Notification **shall** occur prior to implementation of the BCR into the baseline and annotate the type of change, Control Account Manager, Control Account impacted, identification of effects, the value of the change (if applicable), internal Contractor approval, and the accounting month for which the BCR will be implemented.

1.2.6 Estimate at Completion

Once the PMB has been established and approved by the Government, a comprehensive Estimate at Completion **shall** be developed by the Contractor as part of the recurring detailed budget and planning process.

1.2.7 Project Management Plan

The Contractor **shall** perform to and maintain a Project Management Plan (PMP) in accordance with CDRL PM-7.

1.3 Risk Management

The Contractor **shall** establish and maintain a comprehensive risk management program that meets the requirements of Contractor Risk Management Plans and the NEON Program Continuous Risk Management Plan (TBS). All CCB approved, authorized, and funded risk burn-down (mitigation) activities **shall** be integrated into the IMS. The Contractor **shall** invite the Government to attend Contractor Risk Management Board meetings. The Contractor **shall** document, and report all identified risks in accordance with the program's approved Risk Management Plan (RMP) (CDRL PM-8) and the LPD RMP (470-LEO-PLAN-00009). The Contractor **shall** address identified risk areas at program status reviews. The Contractor **shall** maintain a Risk List throughout the program life cycle and add mitigation steps to the IMS, which **shall** be made available to the Government upon request.

1.4 Configuration Management

The Contractor **shall** perform configuration management (CM) in accordance with the Configuration Management Plan, in accordance with CDRL PM-6. The Contractor **shall** maintain and employ a configuration management system to demonstrate that the configuration

item conforms to current approved documentation. The Contractor **shall** maintain configuration of flight hardware, software, electrical ground support equipment (EGSE) and mechanical ground support equipment (MGSE) as well as any equipment which mates to flight hardware throughout all phases of assembly and test. The Software Configuration Management Plan (CDRL SW-3) is discussed in Section 4.3.2.

The Contractor **shall** perform and document configuration verification as parts and assemblies are incorporated into higher-level assemblies and at major program milestones (in accordance with GSFC-STD-1001).

The Contractor **shall** submit a waiver or deviation request for any non-compliance to any requirement. Any flight item that is found to be non-compliant may be re-worked or replaced with a compliant item.

The Contractor **shall** prepare and provide, as a minimum, the following configuration control documentation:

- a) CCB status reported at the Monthly Program Management Review in accordance with CDRL PM-1.
- b) Engineering Drawings in accordance with CDRL SE-3.
- c) The Configuration Item Identification List and the Computer Software Configuration Items (CSCI) in accordance with CDRL RE-8.

1.4.1 Configuration/Change Control Board

The Contractor **shall** be a member of the SMBA GSFC CCB and required to participate in the review of project-level documents which are applicable to the Instrument Contractor (i.e., plans, specs, support of ICD updates, satellite integration and test (I&T) documents, etc.). The Government **shall** be invited to all Contractor Instrument CCBs having the right to participate at its discretion. Government approval is required prior to the implementation of any Class I change. For Class I and II definitions see 470-LEO-PROC-00010, LPD Configuration Management Procedure. Class I changes are modifications affecting the following:

- a) Contractually specified form, fit and function.
- b) Government Furnished Equipment.
- c) Cost to the Government.
- d) Contractual deliveries.
- e) Schedule Contract milestones.
- f) Interchangeability of any item/unit that has already been delivered (excludes launched hardware).

All other changes are considered Class II or lower and may be incorporated without Government authorization. The Contractor **shall** notify the Government of change classifications and provide all reasonable evidence required in support of change classifications. The Government reserves the right to upgrade any change classification to Class I if the Government determines the change to be improperly classified according to the Class I/Class II definitions above.

1.4.1.1 Design Changes

The Contractor **shall** document and control engineering changes as either Class I or Class II. CCRs **shall** be forwarded to the Government in accordance with CDRL SE-1, Engineering Change Requests.

1.4.1.2 Drawings and Drawing List

The Contractor **shall** prepare and maintain a complete set of flight equipment drawings and drawings for GSE. The drawings **shall** be maintained at the Contractor's facility. The Contractor **shall** make available to the Government drawings and an Indentured Drawing List for each hardware type upon request.

1.5 Reviews and Meetings

The Contractor **shall** conduct, host, prepare, and present instrument reviews and provide review packages in accordance with the stated CDRL/Data Item Description (NS1-SMBA-CDRL-0001) for each sensor Instrument as shown in Table 1.

A top-level overview of the reviews to be conducted is provided in Table 1, below. Further details for each review can be found in the subsequent sections of this SOW and the data pack requirements are provided in the CDRL.

Reviews **shall** be considered complete and approved by the Government when all outstanding action items are closed or a detailed plan for closure is submitted by the Contractor and approved by the Government.

Table 1-1. Top-Level Instrument Reviews

Review	Applicability	CDRL Reference	Chair
Integrated Baseline Review (IBR)	FM1	RE-TBS	Government
System Requirements Review (SRR)	FM1*	RE-TBS	Government
System Definition Review (SDR)	FM1*	RE-TBS	Government
Preliminary Design Review (PDR)	FM1*	RE-TBS	Government
ETU Pre-Build Review	ETU	RE-TBS	Contractor
ETU Test Summary Review	ETU	RE-TBS	Contractor
Critical Design Review (CDR)	FM1*	RE-TBS	Government

Review	Applicability	CDRL Reference	Chair
Manufacturing Readiness Review (MRR)	All Flight Instruments		Contractor
Pre-Environmental Review (PER)	All Flight Instruments	RE-TBS	Government
Pre-Ship Review (PSR)	All Flight Instruments	RE-TBS	Government

***If any changes are made to the instrument (design, processes, procedures, etc.), this review may be required at the Government's discretion, depending on the magnitude of the changes.**

Unless stipulated otherwise or agreed to by the Government, the Contractor **shall** hold all Instrument Project Reviews and meetings at the Contractor's facility.

All reviews and meetings **shall** provide a virtual attendance option.

1.5.1 Contractor-Led Meetings and Reviews

The Contractor **shall** implement a thorough technical peer review process of component and subsystem designs, conduct independent technical assessments of instrument activities, and hold formal design reviews. The reviews listed in this section should not be considered a complete set of reviews for the Contractor's program. Additional reviews that the Contractor deems necessary to successfully execute the program will be conducted at the Contractor's discretion.

The Contractor **shall** develop and apply a process for capturing and responding to action items assigned during Contractor-led meetings and reviews. The Contractor **shall** record and deliver meeting minutes for the meetings and reviews called out in this section.

1.5.1.1 Kickoff Meeting

The Contractor **shall** hold a program kickoff meeting approximately two weeks after the Contract award.

The kickoff meeting agenda **shall** be developed jointly with the Government and should include:

- An overview and introduction to the organization and personnel (both vendor and Government teams).
- A top-level look at the total program schedule with a more detailed look at the near-term schedule (between the kickoff and Integrated Baseline Review (IBR)).
- A summary of regularly scheduled meetings and a tentative meeting schedule for the first 6 months (PMRs, tabletop reviews, peer reviews, project milestone reviews, etc.).
- An overview of the instrument design concept.
- An overview of any technology development work.
- An overview of current risk database and risk management approach.
- Overview of configuration management approach.
- Overview of Safety & Mission Assurance (SMA) and quality processes.
- Overview of vendor facilities (tour, if possible).

The kickoff meeting will nominally be held at the vendor facility. Assumed duration is one day.

1.5.1.2 Program Management Review

The Contractor **shall** provide for monthly PMRs in accordance with CDRL PM-1. These reviews will generally be held at the Contractor's facility, but may be held at the Government's facility, with mutual agreement between the Government and Contractor.

The PMR **shall** be designated to provide a management-level review of technical, schedule, and programmatic activities. A PMR will typically consist of a meeting to review formal status and splinter meetings to address special topics to be mutually determined prior to each review. The addition of special topics will be coordinated between the Government and Contractor Managers.

Up to twice a year, the Government **will** coordinate an integrated PMR where the Instrument Contractor and the Spacecraft Contractor **shall** present status and issues. This review **will** be in lieu of the monthly PMR for that month. The Contractor **shall** plan to travel to each combined review through delivery. The integrated PMR will rotate between the SMBA Instrument Contractor's facility and the Spacecraft Contractor's facility. The semi-annual PMR may be planned periodically at GSFC in Greenbelt, Maryland.

1.5.1.3 Integrated Baseline Review

The Contractor **shall** conduct an IBR to assess and validate the PMB in accordance with CDRL RE-6. The Contractor **shall** support a formal IBR conducted by the Government with Government participation. This review will be established by the Government as early as practicable and conducted in accordance with NASA requirements within 180 days of Contract award or formal Contract definitization. Additional baseline reviews may be required when substantial changes are made to the baseline (e.g., subsequent detailed planning, including routine rolling wave planning and program re-planning resulting from incorporation of significant Contract modifications). The IBR will be tailored to meet the needs of the NEON Program.

The Government will coordinate the timeframe and scope of the IBR with the Contractor. The IBR will ensure that the PMB captures the entire scope of work, is consistent with the Contract schedule requirements and has an adequate resource plan. The IBR team **shall** include Government personnel.

1.5.1.4 Engineering Test Unit Pre-Build Review

The Contractor **shall** hold an Engineering Test Unit (ETU) Pre-Build Review to get Government agreement for the ETU Program – the scope and fidelity of the ETU and any demonstrators (e.g. breadboards, simulators, etc.), what testing will be done, and an attempt to define priorities ahead of time to aid in quicker decision making should there be cost, schedule, procurement, or facilities challenges that require an adjustment to the scope or fidelity of the ETU.

The Contractor **shall** deliver the ETU Pre-Build Review data package in accordance with CDRL RE-12.

1.5.1.5 ETU Test Summary Review

The Contractor **shall** hold an ETU Test Summary Review to present the actual testing conducted compared to the plan, the results and any lessons of that testing, and any subsequent changes, either to the ETU and/or the flight design.

The ETU Test Summary Review may be combined with the Critical Design Review (CDR) if mutually agreed by the Contractor and Government.

The Contractor **shall** deliver the ETU Test Summary Review data package in accordance with CDRL RE-13.

1.5.1.6 Design Modification Reviews

Whenever there is a design change to the flight instrument that the Government feels cannot be adequately reviewed by other means, the Contractor **shall** hold a Design Modification Review (DMR) to review these changes at the Government's request. The Contractor **shall** include Government representatives on the review board.

The Contractor **shall** deliver a DMR Data Package in accordance with CDRL RE-9.

1.5.1.7 Weekly Meetings

In order to track status, coordinate activities, and resolve issues, the Contractor **shall** support weekly meetings and telecons with the Government at mutually agreed to times. At a minimum, the weekly meetings should include: 1) a meeting to review status, work completed in the prior week, and a look ahead; 2) a technical issues meeting to cover technical issues as mutually agreed by the Contractor and Government; 3) a brief weekly financial and business meeting to review staffing, spending and funding, and special studies task status and hours.

1.5.1.8 Engineering Peer Reviews

The Contractor **shall** define and implement a series of Engineering Peer Reviews (EPR) for the hardware and software subsystems of the instrument commensurate with the scope, complexity, and acceptable risk of the product (CDRL PM-3). The purpose of an EPR is to reduce risk through expert knowledge infusion, confirmation of approach, and specific recommendations. The Contractor **shall** chair and host EPRs at the Contractor's facilities with both Government decisional and non-decisional participation on the review panels. The Contractor **shall** conduct multiple peer reviews, as appropriate, over the life cycle of each subsystem and component with content consistent with the evolving design and development.

At a minimum, the Contractor **shall** complete a comprehensive series of subsystem or lower-level peer reviews prior to any instrument design reviews. Successful completion of these EPRs and resolution of associated technical issues and actions is an important aspect of entry criteria in the formal review process. The Contractor **shall** use EPRs for the focused evaluation of concepts, designs, plans, and processes associated with combinations of subsystems and system functions and performance that cross traditional subsystem or discipline boundaries.

The Contractor **shall** conduct peer reviews as defined in the Peer Review Plan in accordance with CDRL PM-3 and document the peer reviews in accordance with CDRL RE-7, Engineering Peer Review Data Packages.

1.5.1.9 Manufacturing Readiness Review

The Contractor **shall** conduct a Manufacturing Readiness Review (MRR) with the Government and its representatives in attendance to be conducted after flight hardware procurement and prior to the start of flight hardware manufacturing. The purpose of the MRR is to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that all prerequisites have been met, all necessary paperwork is correct and complete, and the final and correct material is released to manufacturing and that the manufacturing planning is complete and reflects the final configuration. The MRR data package **shall** include all documentation required to manufacture hardware in compliance with the Contract requirements and submitted in accordance with CDRL RE-8. The MRR **shall** be considered complete and approved by the Government when all outstanding action items are closed or a detailed plan for closure is submitted by the Contractor and approved by the Government.

MRRs **shall** be conducted for lower-level assemblies and LTS as documented in the Government-approved PMP.

1.5.1.10 Test Readiness Reviews

The Contractor **shall** conduct Test Readiness Reviews (TRR) immediately prior to the start of any test of critical sub-assemblies, subsystems, and the instrument.

Unless stipulated otherwise by the Government, the Contractor **shall** include Government personnel in TRRs and in the test procedure review process and provide a copy of the final procedure prior to the review.

The Contractor **shall** deliver TRR Data Packages in accordance with CDRL RE-11.

1.5.1.11 Test Data Reviews

The Contractor **shall** conduct Test Data Reviews (TDR).

Unless stipulated otherwise by the Government, the Contractor **shall** include Government personnel in TDRs. The Government **shall** have voting rights in all consent-to-break decisions made based on the TDR.

The Contractor **shall** conduct TDRs for data collected from instrument critical sub-assembly, subcontracted subassembly, and instrument, as well as prior to the shipment of deliverable GSE.

The Contractor **shall** conduct a TDR of the interim data during the qualification or acceptance test process of the instrument prior to the breakdown of any test setup (e.g., vibration testing, Electromagnetic Capability (EMC)/Electromagnetic Interference (EMI) testing, thermal vacuum testing, etc.) to ensure successful collection of required data, and prior to the Pre-Ship Review (PSR), to determine readiness.

1.5.1.12 Software Reviews

The Contractor **shall** plan and conduct software reviews in accordance with the Instrument Software Development Plan, CDRL SW-2.

Unless stipulated otherwise by the Government, the Contractor **shall** include Government personnel in software reviews.

The Contractor **shall** conduct the following software reviews:

- a) Software Requirements Specification Review/Walkthrough
- b) Software Preliminary Design Review
- c) Software Critical Design Review
- d) Software Test Readiness Review
- e) Software Qualification Review

The Contractor **shall** provide data packages for the following reviews:

- a) Software Test Readiness Review (SWTRR) Data Package in accordance with CDRL SW-10.
- b) Software Acceptance Review (SWAR) Data Package in accordance with CDRL SW-11.

The Contractor **shall** conduct software peer reviews.

The Contractor **shall** document the scope and process for conducting software peer reviews in the Instrument Software Development Plan in accordance with CDRL SW-2.

1.5.1.13 Technical Interchange Meetings and Other Reviews and Meetings

A Technical Interchange Meeting (TIM) is an informal meeting between the Contractor and Government representatives. For example, to reach understanding of an operation or analysis, presentation of test results, discuss planned interface changes, plan for an upcoming test, etc. TIMs typically are held at the Contractor's facility and typically run no more than two days. Actions are tracked by the TIM organizer.

The Contractor **shall** conduct technical/management interchange meetings as needed to discuss and resolve any technical and/or managerial issues or concerns.

Within 1 week of concluding the TIM, the Contractor **shall** submit the briefing materials, the attendees, action items, and significant decisions to the Government.

Per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (TBS) within Contract TBS, Clause TBS, Attachment TBS, the Contractor **shall** invite the Government to participate in the Contractor's Failure Review Board (FRB) and Material Review Board (MRB). Mission Assurance boards (MRB, FRB, Parts Control Board (PCB), Materials and Processes Control Board) are prescribed in the IMAR document.

In addition to the meetings and reviews described above, the Contractor **shall** support routine informal meetings and telecons with the Government. Weekly meetings or telecons between the

Government and the Contractor **shall** be supported by the Contractor at mutually agreed to times in order to coordinate activities and resolve issues.

1.5.1.14 Subcontract and Subsystem Reviews

The Contractor **shall** plan and conduct subcontract and subsystem reviews in accordance with the Project Management Plan, CDRL PM-7.

Unless stipulated otherwise by the Government, the Contractor **shall** include Government personnel in subcontract and subsystem reviews.

The Contractor **shall** conduct Preliminary Design Reviews (PDR) and CDRs for all major subsystems whether the subsystems are provided by the Contractor or by a subcontractor.

Subcontract and subsystem reviews **shall** be conducted by the Contractor. The review board **shall** be chaired and staffed by Contractor-selected independent reviewers for the hardware under review.

For subcontractor and subsystem reviews, the Contractor **shall** document the minutes.

The Contractor **shall** submit Subsystem/Subcontract Review Data Packages in accordance with CDRL (TBD).

1.5.2 Government-Chaired Instrument Reviews

The Government will chair the major milestone reviews in the instrument life cycle where information is formally presented to a panel of Government experts and external reviewers. These reviews will typically involve 10 to 50 Government representatives. The Contractor **shall** determine the optimum number of consecutive days to perform each of these reviews to meet the requirements in GSFC-STD-1001, but not to exceed four days. The Contractor **shall** conduct dry runs of all System-level Reviews with the Government approximately two to four weeks in advance of the reviews, or as agreed to with the Government. These dry runs may be conducted either face-to-face or by electronic means. Formal action items are logged and tracked by the Government Review Team.

All major reviews will be convened, and review boards appointed and chaired by the Government. The Contractor **shall** demonstrate compliance with the review success criteria of GSFC-STD-1001, Criteria for Program Flight Critical Milestone Reviews, as applicable to spaceflight instruments. The Contractor **shall** work with the Government program and the review team for an additional day following all reviews to discuss and address issues raised and actions assigned at the reviews.

The Contractor **shall** develop and apply a process for capturing and responding to action items assigned by the Government. Reviews are not complete until actions are complete or a detailed plan for closure is submitted by the Contractor and approved by the Government.

Where the Government determines an additional review is required, a “delta” review **shall** be conducted. This “delta” review **shall** meet the requirements of the appropriate System-level review (CDR, etc.) as defined herein and in the applicable CDRL. Once the Government has

decided a “delta” review is necessary, the Government will formally inform the Contractor of the tailored scope of the delta review and the deadline to hold the review.

1.5.2.1 System Requirements/Definition Review

The Contractor **shall** conduct a System Requirements Review (SRR)/System Definition Review (SDR) to ensure proper and complete requirements definition and flowdown from the Level 3 requirements levied by the Government to lower levels.

The Contractor **shall** deliver an SRR/SDR Data Package in accordance with CDRL RE-1.

1.5.2.2 Preliminary Design Review

The Contractor **shall** conduct a PDR at the conclusion of the preliminary design efforts per GSFC-STD-1001. The Contractor may combine the PDR with the engineering development model Pre-Build Review pending agreement from the Government IM and respective review chairpersons.

The Contractor **shall** deliver a PDR Data Package in accordance with CDRL RE-2.

1.5.2.3 Critical Design Review

The Contractor **shall** conduct a CDR prior to the start of procurement and/or manufacture of the Protoflight Model hardware unless the Contractor requests and the Government concurs with an earlier start. The Contractor **shall** obtain Government concurrence to purchase any materials prior to the CDR.

The Contractor **shall** deliver a CDR Data Package in accordance with CDRL RE-3.

1.5.2.4 Pre-Environmental Review – Each Instrument

The Contractor **shall** conduct a Pre-Environmental Review (PER) prior to the start of environmental testing of the ETU and each Flight Model (FM) instrument to establish the readiness of the system and to evaluate the environmental test plans and procedures.

The Contractor **shall** deliver a PER Data Package in accordance with CDRL RE-4.

1.5.2.5 Pre-Shipment Review – Each Instrument

The Contractor **shall** conduct a pre-ship review (PSR) at the conclusion of the integration and test effort prior to shipment and/or storage of each SMBA FM. If the instrument is stored for a period greater than six months prior to shipment, the Contractor **shall** prepare an abbreviated (half-day) delta-PSR immediately prior to shipment to review logistical shipping details as well as any additional work or testing performed on the instrument since the PSR.

The Contractor **shall** deliver a PSR Data Package, in accordance with CDRL RE-5.

1.5.3 Higher-Level Reviews

The Contractor **shall** provide support to other reviews as required by the NEON Program. Once the instrument is delivered, the Contractor **shall** support all reviews necessary for the successful integration, test, launch, and operation of this instrument as defined in Section 6 of this SOW.

1.5.3.1 Satellite-level Reviews

The Contractor **shall** provide technical support to all Satellite-level reviews, including Satellite integration and test reviews, conducted at the Spacecraft Contractor's facility.

The Contractor **shall** support the Satellite Mission-level reviews by presentations and/or documentation; for example, by providing and/or presenting instrument overview or design summary, test, and data evaluation summaries, etc., in accordance with CDRL RE-14, "Inputs to Mission-Level Reviews."

The Contractor **shall** attend all Satellite-level Reviews including but not necessarily limited to:

- a) Integration Readiness Review (1 day for each FM)
- b) Satellite Pre-environmental Review (3 days for each FM)
- c) Satellite Pre-storage Review (3 days for each FM)
- d) Satellite Pre-ship Review (3 days for each FM)

1.5.3.2 Program-Level Reviews

The Contractor **shall** support the Program-level Reviews by presentations and/or documentation, for example, an instrument overview or design summary, test, and data evaluation summaries, etc.

The Contractor **shall** support the following Program-level Reviews:

- a) Mission SDR (3 days)
- b) Mission PDR (3 days)
- c) Mission CDR (3 days)
- d) System Integration Review (3 days for each FM)

Any inputs **shall** be provided in accordance with CDRL RE-16, "Inputs to Higher Level Reviews."

1.5.3.3 Operations Reviews – Each Flight Model

The Contractor **shall** support the Operations Reviews for each FM by presentations and documentation; for example, by providing and/or presenting instrument overview or design summary, test, and data evaluation summaries, etc., in accordance with CDRL RE-16, "Inputs to Higher Level Reviews."

The Contractor **shall** support the following Operations Reviews for each FM at the Satellite Operations Facility :

- a) Mission Operations Review (MOR) (3 days)
- b) Flight Operations Review (3 days)
- c) Operational Readiness Review (ORR) (3 days)
- d) Post-Launch Assessment Review (2 days)

1.6 Data Retention

The Contractor **shall** retain those data necessary to provide a technical and program database of significant decisions, plans, and the execution thereof as well as those data providing design,

manufacturing, and test history of all deliverable items for the period of 10 years from the date of the Contract final payment/settlement. Data to be retained include the following:

- a) Part and material procurement: Test, screening, and inspection data, including nonconformance reports.
- b) Subassembly and higher assembly: Fabrication, assembly, test, and inspection data.
- c) Shop travelers (including travelers at lower tier suppliers).
- d) Kitting/traceability records.
- e) Executed test procedures.
- f) Listing of test equipment used.
- g) Test and calibration data including nonconformance reports.
- h) GSE certification, calibration of test equipment.
- i) As-built/as-designed verifications.
- j) Operating time logs.
- k) Total test time for each environment.
- l) End Item Data Packages.
- m) Design review data and packages.
- n) Drawings.
- o) Contractor Parts, Materials, and Processes Control Authority records of all performed testing:
 - Qualification and acceptance test data.
 - Destructive physical analysis reports with samples.
 - Radiation hardness assurance test data.
- p) Computer Aided Design data files that capture the mechanical design of the instrument and associated GSE.
- q) Photographic documentation.
- r) Models used to perform analyses for delivered CDRLs (e.g., thermal, optical, contamination, General Reflector Antenna Analysis Software Package).
- s) Environmental records.
- t) Retain numeric data in digital format.
- u) Retain or archive software lookup tables.

2 SYSTEMS ENGINEERING

The Contractor **shall** perform systems engineering to support all instrument-related activities during all stages of development. The systems engineering effort **shall** include, but is not limited to, analyses of technical requirements, functional and performance allocation of derived requirements, traceability, definition and maintenance of all interfaces, instrument design and verification of all defined, allocated, and derived requirements, systems analyses and special studies as required, risk management support, and tradeoff analyses. This includes but is not limited to the following specific activities:

- a) Providing technical direction and oversight throughout all phases of the program.
- b) Leading, supporting and/or preparing documentation for all peer reviews as well as those reviews defined in Section 1.5.
- c) Performing system studies, trades, and risk assessments necessary to develop the instrument design.
- d) Performing coordination, studies and analyses required to interface the instrument to the spacecraft (mechanical interface control document (MICD), ICD), including instrument specific support to ground system requirements development and operations.
- e) Performing systems engineering and analysis in support of instrument tests at the satellite-level (testing or work done on SMBA while it is integrated with the spacecraft bus) and launch site.
- f) Supporting system-level technical interface meetings, including technical issue resolution, performance verification plan buy-offs, pending configuration change requests, CDRL data submission review/approval status, test data review, anomaly resolution activities, and test support planning.
- g) Developing a Systems Engineering Management Plan (CDRL SE-18).
- h) Provide a Firmware Design Data Package in accordance with CDRL RE-10.

2.1 Deviations and Waivers

Requests for deviations or waivers to the Government's requirements **shall** be submitted to the Government's Contracting Officer for Government approval and documented in CDRL SE-1.

2.2 Requirements Analyses, Derivations, and Allocations

The Contractor **shall** provide the definition, allocation, derivation, and traceability of system and subsystem requirements and the verification approach.

The Contractor **shall** conduct complete analyses and simulations in support of technical requirements compliance demonstrations to fully establish, define, maintain, and control allocations for all required performance and design parameters, including:

- a) Flow-down and traceability of Instrument requirements from SMBA requirements per CDRL RE-1 Specification Tree.
- b) Functional and performance allocations and derivations in accordance with CDRL RE-1.
- c) Developing the Instrument Concept of Operations in accordance with CDRL SE-8.
- d) Preparing the Instrument Mass Properties Reports in accordance with CDRL SE-23.
- e) Developing the Beginning of Life (BOL) design performance requirements based on the PRD End of Life (EOL) requirements and documenting the supporting analyses in CDRL SE-6, Instrument Performance Specification.

If a Model-based systems engineering approach is used to support the requirements, design, analysis, verification, and validation associated with the development of the SMBA instrument models, any analysis and supporting documents (CDRL SE-7) **shall** be provided virtually.

2.2.1 Technical Performance Metrics

The Contractor **shall** implement and manage Technical Performance Metrics (TPM), comprised of key performance requirements for the end item product, including tracking and reporting TPM status and changes. The Contractor **shall** prepare and submit TPM status detailing current estimated value, requirement, margin, trend, and corrective action if required, in accordance with CDRL PM-1. Assessment updates **shall** be performed quarterly and at the completion of significant design and development tasks. If there are no changes to report reference can be made to the last reported TPM baseline.

2.2.2 Instrument Interface

The Contractor **shall** support the development of bilateral instrument documents (Government to Contractor, Spacecraft to Instrument), as well as analyses and models which define the details of the instrument-to-spacecraft interface, instrument accommodation, and operating procedures and information for the instrument.

2.2.2.1 ICD & Mechanical ICD

The Instrument Contractor **shall** develop and deliver an Instrument ICD (SE-9) and MICD (SE-10) that includes the instrument's interface requirements that will be provided to the spacecraft Contractor to form the basis of the instrument-to-spacecraft ICD/MICD. The Contractor **shall** collaboratively work with the Government and the spacecraft Contractor to review and maintain accurate instrument-to-spacecraft ICD(s) and MICD(s) and recommend corresponding and/or similar changes to the PRD. The Contractor **shall** verify ICD requirements in accordance with CDRL SE-4.

2.2.2.2 Giver/Receiver

The Contractor **shall** collaboratively support the development, maintenance, and review of the SMBA Giver/Receiver list of CDRLs and other items (such as, but not limited to, documents, hardware, and software) to be exchanged between the Contractor and other organizations. The Contractor **shall** provide the items documented in the Giver/Receiver list per the schedules documented and agreed to therein. The Giver/Receiver list **shall** document both items to be delivered to outside organizations (Givers) and items to be delivered from outside organizations (Receivers).

The Giver/Receiver List **shall** be delivered as CDRL PM-9.

2.3 Contamination Control

The Contractor **shall** incorporate contamination control into the instrument design that assures mission success per IEST-STD-CC1246D. The instrument **shall** be maintained at a cleanliness level as defined in SMBA Satellite Contamination/Cleanliness Control Plan. The Instrument **shall** be cleaned prior to its use, shipment, and integration onto the spacecraft. Vent locations and paths **shall** be defined in the MICD. The Contractor **shall** monitor the instrument for contamination effects and validate instrument contamination controls. The Contractor **shall**

document contamination analyses as specified in the instrument's Contamination Control Plan (MA-30) and analysis reports (CDRL SE-7f).

Bake out **shall** be done prior to delivery, such that contamination requirements are met prior to delivery to the spacecraft facility to minimize contamination of other contamination sensitive components/subsystems on the space vehicle during storage and on orbit.

Multi-layer insulation blankets, harnesses and cables, and electronic boxes/modules to be included in thermal-vacuum testing **shall** be baked out separately from the instrument.

Purge port: If there is a conditional purge, it **shall** be delivered via a purge port rather than a feed through a bag.

For long term storage there **shall** be a purge port that remains capped for I&T. The purpose would be to provide flow through the instrument volume to prevent the build-up of corrosives due to offgassing.

If the Instrument does not have a purge port accessible after delivery and integration, the Contractor **shall** conduct a contamination study to determine the best value approach to protecting the instrument from contamination sources that exist through launch and deliver the data, results, and recommendations to the Government for disposition.

Powered operations while bagged: The Contractor **shall** verify whether the Instrument can operate while bagged and define how long it can operate while bagged in the case of thermal constraints.

The Contractor **shall** supply any instrument bags needed after delivery to the spacecraft.

Post-delivery support: The Contractor **shall** be responsible for instrument cleaning and sampling post-delivery through encapsulation.

The MGSE **shall** be delivered in a state that prevents particulate contamination.

2.4 Design Engineering

The Contractor **shall** design and deliver an instrument that meets all requirements, requirements in the PRD, IMAR, CDRL, ICD and MICD. The Contractor **shall** develop and deliver the Instrument Performance Specification in accordance with CDRL SE-6. The Contractor **shall** provide all necessary personnel, facilities, services, and materials to design the instrument in accordance with its design specifications. The Contractor **shall** make available for review by the Government design specifications and drawings for all subcontracted and purchased items. Responsibility for the design, manufacturability, fabrication, testability, and associated functions belongs to and **shall** remain solely with the Contractor.

The Contractor **shall** conduct subsystem requirements and design EPRs for, at a minimum, the following subsystems prior to milestone reviews:

- a) Electrical Systems in accordance with SE-11 and RE-7.
- b) Antenna in accordance with RE-7.

- c) Scan Drive and electronics in accordance with RE-7.
- d) Receivers in accordance with RE-7.
- e) Thermal in accordance with RE-7.
- f) Structural in accordance with RE-7.
- g) Software in accordance SW-5 and RE-7.

2.5 Design, Analysis, and Verification Documentation

The Contractor **shall** perform and document all analyses of the data and information from the design, qualification testing, acceptance testing, compatibility testing, and on-orbit testing of the Contractor's hardware and software which are required to ensure that the instrument will meet its specifications and objectives.

2.5.1 Design Analyses

The Contractor **shall** perform the necessary systems engineering analyses to support the satisfaction of the Contract requirements in accordance with CDRL SE-7 and CV-4, including the following:

- a) Radiometric Math Model with uncertainties defined, explained, and quantified.
- b) Perform a flow-down of the performance requirements to meet its requirements with sufficient margin.
- c) Pointing knowledge and error budget of the entire instrument.
- d) Data system throughput, storage, and any playback analyses.
- e) Antenna cross-polarization characterization with respect to scan angle.
- f) Antenna alignment design, tolerances, and error budget.

All analyses performed by the Contractor under this Contract **shall** be made available to the Government.

Design analyses **shall** be submitted in accordance with CDRL SE-7 in support of the design reviews. The Contractor may submit existing analyses for any unit type for which space qualification has been demonstrated to comparable requirements. Contractor format and analyses configuration is acceptable.

2.5.1.1 Structural/Dynamic Analyses and Verification Model and Plan

The Contractor **shall** perform comprehensive structural and mechanical analysis and deliver a report documenting the analysis, develop and deliver Structural and Dynamic Models, and a Model Verification Plan in accordance with CDRL SE-7a. The Contractor **shall** get concurrence from the Government on whether a test is required to correlate the models with the structural test results to verify the accuracy.

2.5.1.2 Thermal Analyses and Modeling

The Contractor **shall** provide documented verification that the hardware meets all thermal design requirements for the program. The Contractor **shall** develop and deliver a comprehensive thermal analysis, a Thermal Analysis Report documenting the analysis, and Thermal Models in accordance with CDRL SE-7b. Thermal analyses **shall** be performed to demonstrate that test configurations and associated ground support equipment (e.g., cold plates, chillers, chambers, thermal targets) have sufficient control authority to thermal vacuum cycle the flight hardware to protoflight levels and simulate mission environments during thermal balance tests. The detailed

analysis **shall** demonstrate that all parts meet established thermal design criteria, based upon program de-rating or electrical performance requirements.

The instrument Contractor **shall** define temperature limits of components as well as test tolerances during PDR, in coordination with the Government responsible personnel.

Thermal verification tests **shall** be configured so that heat pipes are in gravity neutral or reflux orientations to ensure operation of these devices during thermal testing.

Full and Reduced Thermal Math Models (CDRL SE-7b) of the instrument **shall** be delivered to the Government for use in designing the spacecraft-instrument interfaces. The detailed instrument models will be combined with the analytical models of the spacecraft to create a reduced system-level thermal model of the satellite. Only the instrument Reduced Thermal Model (RTM) will be delivered to the Spacecraft Contractor by the Government. The Government will deliver the spacecraft RTM to the instrument Contractor. Only the Government will have and run the integrated full models.

If there are any special requirements such as special temperature control, heater power, thermal distortions, etc., which are pertinent to the design, these considerations **shall** be identified and included in the analysis.

The Contractor **shall** use Thermal Desktop PC-based software for thermal analyses and modeling.

2.5.1.3 Stress Analysis

The Contractor **shall** perform a detailed stress analysis and margins of safety calculations for all components of the instrument and provide a Stress Analysis Report documenting the analysis in accordance with CDRL SE-7c.

Shock and Jitter Analyses: The analyses **shall** demonstrate that the hardware meets the applicable shock and jitter requirements defined in the equipment specification. Shock and jitter analyses **shall** be submitted in accordance with CDRL SE-7e and SE-7g, respectively.

2.5.1.4 Electromagnetic Capability/Electromagnetic Interference Analysis

The Contractor **shall** perform EMC/EMI analysis and/or test in accordance with the SMBA Performance Requirements Document (470-SMBA-00435) and MIL-STD-461 to verify the hardware meets the performance requirements of the equipment specification and provide an EMC/EMI Analysis Report documenting the analysis in accordance with CDRL SE-7h.

2.5.1.5 Radiation Hardness and Survivability (H&S) Analyses

The Contractor **shall** perform BOL and EOL H&S analyses to verify the equipment meets the requirements in accordance with CDRL SE-7d.

Radiation Analysis Report: This analysis and report **shall** include radiation effects/degradations on components, coatings, and on electronic performance; Dose Mapping analysis; Single Event Effects analysis; Displacement Damage analysis; parts total dose analysis; and test data supporting these analyses. Transient radiation effects that are not distinguishable from noise in

the instrument output **shall** be included as a noise component in signal-to-noise ratio analyses and compliance evaluations. Transient radiation effects on temperature measurements used in any calibration **shall** be analyzed, including impacts on radiometric calibration.

2.5.1.6 Radiometric Math Model/Sensor Performance Math Model – Design and Development Documentation

The Contractor **shall** develop, deliver, and maintain a Radiometric Math Model/Sensor Performance Math Model (SPMM) in accordance with CDRL CV-4. It **shall** include assumptions and assessments of assumptions as to their fidelity. The SPMM is expected to supply and support system performance predictions.

2.5.1.7 Radio Frequency Interference Analysis

RFI analysis in accordance with SE-7j.

2.5.1.8 Hyperspectral Capability Analysis

Hyperspectral Capability Analysis in accordance with SE-7i.

2.5.1.9 Intercalibration Analysis

Intercalibration Analysis in accordance with CV-7.

2.5.2 Trending

The Contractor **shall** establish and implement a method for trending test data for all critical functions across missions through the life of the program in accordance with CDRL SE-2.

The trending method may differ between instrument level testing and satellite level. The Contractor **shall** coordinate with the Government the selected list of parameters to be trended in accordance with Trending Plan, Analysis Reports, and Operations Log (CDRL SE-2). The Contractor **shall** monitor selected parameters for trends starting at the beginning of instrument electrical integration and continuing during the system integration and test phases through the on-orbit commissioning phase. The Contractor **shall** support trending of data across missions through the life of the program. The Contractor **shall** analyze the trended data for indications of anomalous conditions and for possible performance or reliability degradation. The Contractor **shall** provide Trending Reports and operating hours in accordance with the Trend Analysis, Reports, and Operations Log. The Contractor **shall** coordinate with the Government the list of parameters to be trended on orbit. The Contractor **shall** support the development and validation of any Mission Operations Element operational trending system by participation in telecons as well as review and comment on documentation.

The Contractor **shall** review and report all trending results from every phase of the development with the Government.

2.6 Special Studies

The Contractor **shall** conduct, in addition to the requirements specified in this document and the Contract, additional engineering studies, tests, technical analyses, reviews of test results, design modifications, and tasks relating to the development, implementation, characterization, and operation of the instrument, as authorized by the Government and in accordance with Contract Section H Clause, Special Task Support Requirements. Each task will be initiated by written

direction from the Government Contracting Officer. The Government will coordinate with the Contractor to define each task in detail, and establish manpower ceilings, performance schedules, and deliverables.

2.7 Constraints

The Contractor **shall** identify all constraints that pertain to all integration, test, and operational environments, including testing while encapsulated at the launch pad as described in CDRL OO-7, including commands that can potentially damage the spacecraft or instrument, or cause injury to personnel.

3 MISSION ASSURANCE

The Contractor **shall** develop, implement, and maintain a comprehensive mission assurance program which meets the requirements of the SMBA IMAR document (NS1-SMA-REQ-0002), as tailored by SMBA Mission Assurance Compliance Matrix (CDRL MA-1). The Contractor **shall** deliver mission assurance data per the SMBA IMAR, as tailored by SMBA Mission Assurance Compliance Matrix and SMBA CDRL (NS1-SMBA-CDRL-0001. Upon Government acceptance of NS1-SMBA-CDRL-0001, the SMBA IMAR Compliance Matrix will become an applicable document to this Contract.

3.1 Deviations and Waivers

Requests for deviations or waivers to the Government's requirements **shall** be submitted to the Government's Contracting Officer for Government approval per the IMAR, as tailored by SMBA Mission Assurance Compliance Matrix (NS1-SMA-REQ-0002) within Contract TBS, Clause J.1, Attachment N (TBR), and CDRL SE-1.

3.2 Equipment Tagging

All items (non-flight equipment) to be removed prior to flight **shall** be tagged with a Contractor's standard red tag, "Remove Before Flight."

All items to be installed at the launch site, such as instrument flight plugs, **shall** be considered "green tag" (Install Before Flight) items.

4 FABRICATION AND ASSEMBLY

The Contractor **shall** provide all necessary personnel, facilities, services, materials, tooling, and test equipment to fabricate and assemble the Instrument in accordance with its design specifications and Contract requirements. “Fabrication” in this sense also includes procured items.

A Manufacturing (Fabrication and Assembly) Flow Plan (CDRL PM-5), or equivalent, **shall** be maintained and updates made available to the Government upon request.

4.1 Spares

The Contractor **shall** develop, deliver, and implement an approved spare parts program plan for critical flight spares and for critical GSE spares, including spares replenishment, built by the Contractor (e.g. Calibration Test Equipment, target assemblies, and electronics) in accordance with CDRL SE-12.

Once the plan is approved by the Government, the Contractor **shall** procure sufficient spare parts to account for typical attrition during the manufacturing process due to handling, random failure, programming, etc. The spare parts **shall** be qualified, tested, stored, and calibrated to the same standards as the flight instrument articles consistent with their spared level of assembly.

4.2 Software Development

The Contractor **shall** adhere to the NASA Procedural Requirement (NPR) 7150.2, NASA Software Engineering Requirements, as summarized in Appendix A, for the Instrument Flight Software and Instrument Simulator Software for any software changes beyond the SMBA baseline. These requirements are tailored by the Government for Class B software, where software includes commercial-off-the-shelf (COTS), Government-off-the-shelf (GOTS), modified-off-the-shelf (MOTS), reused software, custom software, and firmware and programmable logic devices which utilize a microprocessor; and for Class C software, where software includes ground software for simulations used to test flight software. As part of that adherence, the Contractor **shall** complete an NPR 7150.2 Tailoring Compliance Matrix against those requirements (CDRL SW-1). For firmware and programmable logic devices which utilize a microprocessor at the box-level, the only applicable requirement from NPR 7150.2a is SWE-027. (Software Safety NASA-STD-8719.13 and the Software Assurance NASA-STD-8739.8 requirements still apply for these cases, but those requirements will be verified at the box-level.)

For Field Programmable Gate Array (FPGA) development, if utilized, reference the design guidelines, NASA GSFC 500-PG-8700.2.7 Procedures and Guidelines for Design of Space Flight Field Programmable Gate Arrays and NASA GSFC 500-PG-8700.2.8, FPGA Development Methodology (SE-15 and SE-16).

4.2.1 Software Development Plan

The Contractor **shall** develop and control the instrument flight software in accordance with the Instrument Software Development Plan (SDP) (CDRL SW-2). The Government approved SDP **shall** apply to software developed for this Contract, whether developed by the Contractor or by a third party (i.e., COTS, GOTS, MOTS, subcontractor) and whether developed for flight use or for GSE/simulators.

The Contractor **shall** identify in the SDP software and heritage software reuse plans. The Contractor **shall** perform an analysis of unnecessary and/or unreachable code on the intended flight version of the instrument flight software and produce an Unreachable Code Analysis document to be made available for Government review. The analysis **shall** identify all instances (areas) of unnecessary/unreachable flight code, the general functionality associated with the code, the reason each is intended to be left within the flight load, and the justification (e.g. mitigating action) that explains why the included code does not pose a risk to the mission. The focus will be on technical risk to the long-term mission, not cost.

4.2.2 Software Configuration Management

The Contractor **shall** implement a software CM Plan (CDRL SW-3). This may be combined with a program CM Plan.

4.2.3 Software Requirements Specification

The Contractor **shall** submit an Instrument Software Requirements Specification (SRS) documenting the instrument flight software requirements (CDRL SW-4).

The Contractor **shall** perform all analyses and software systems engineering required to identify and allocate the software requirements (given the system and subsystem requirements). The Contractor **shall** ensure that all requirements are forward and backward traceable between system and software requirements and between software requirements, design, and test.

4.2.4 Software Detailed Design

The Contractor **shall** maintain software design details in accordance with CDRL SW-5 and **shall** deliver an Instrument Software Detailed Design Document (CDRL SW-5).

Instrument Software Development Folders (SDF) and Instrument Software Engineering Notebooks (SEN) **shall** be initiated by the Contractor at the start of software planning and maintained through Contract end. The content and process to be used for the instrument's SDF and SEN **shall** be documented in the SDP. Delivery of these documents to the Government is not required, but they **shall** be made available for audit by the Government at the Contractor's facility. Issues identified **shall** be addressed by the Contractor within 60 days of the finding.

The Contractor **shall** document and maintain instrument GSE/STE Software detailed design and make it available for Government review; the Contractor **shall** also maintain GSE/STE SDFs and GSE/STE SENs initiated at the start of GSE/STE system planning and maintained through Contract end. The content and process to be used for the SDFs and SENs **shall** be documented in the Software Development Facility Specification. Delivery of the GSE/STE SDF and SEN to the Government is not required, but they **shall** be made available for audit by the Government at the Contractor's facility. Issues identified during an audit **shall** be addressed within 60 days.

4.2.5 Instrument Software Development Environment

The Contractor **shall** develop the software development and validation environment for the instrument that supports the development and test of the flight software. It includes host development computer operating systems, high-level language compilers and debuggers, autocode generator software systems, machine language emulators, and test scenarios and

procedures. It includes the software in the instrument test environment simulators that model the detectors and mechanisms. It also includes development support software such as configuration management systems for tracking software and document versions. Documentation of the development environment components **shall** be provided in an Instrument Software Development Facility Specification (CDRL SW-13).

4.2.6 Software Testing

The Contractor **shall** plan and implement a software verification and validation program to ensure that software being developed or maintained satisfies functional and performance requirements defined in the SRS. The Contractor's testing approach and methodology **shall** be documented in the Software Test Plan (CDRL SW-6). This test plan **shall** describe all levels of software testing (unit, component, qualification, validation, and acceptance), the software test organization, the software integration and test program, and simulation, test equipment, and other facility components.

The Contractor **shall** compile Software Test Descriptions, including detailed descriptions of all qualification and acceptance tests; Software Test Procedures (CDRL SW-7), including acceptance criteria for software prior to and after loading; and Software Test Reports (CDRL SW-8), including the results of all qualification and acceptance tests. Test descriptions, procedures, and results **shall** be made available for Government review.

4.2.7 Software Version Description

Instrument flight software and firmware code **shall** be delivered accompanied by a Software Version Description Document (CDRL SW-9) describing the deliverable code and the associated operating procedures. For all firmware media, Programmable Logic Devices and formatted and numbered drawings are required, with assembly numbers assigned to all loaded chips and drawing notes used to identify specific loading and test procedures, markings, etc.

4.2.8 Software User Manual

The Contractor **shall** prepare and deliver a Software User's Guide (CDRL SW-12) describing the installation, build, and operating procedures for the flight software.

4.2.9 Software Reviews

The Contractor **shall** prepare and conduct flight software status reviews as part of the technical status portion of the Monthly Program Management Reviews whenever testing or changes to the flight software baseline are being planned or are occurring.

Software TIMs **shall** be conducted with Government participation.

In addition to the instrument-level EPRs, the Contractor **shall** conduct, as directed by the Government, the following software-specific EPRs:

- Software Requirements EPR.
- Flight software acceptance TRR and deliver a Software TRR Data Package (CDRL SW-10).
- SWAR and deliver a Software Acceptance Review Data Package (CDRL SW-11).

The Government **shall** be invited to attend all Qualification TRR and Formal Qualification Testing.

4.2.10 Software Measures (Metrics)

The Contractor **shall** acquire software measures (metrics) as defined in the SDP from any subcontractors or team members and provide them to the Government both as raw data and in graphical form as part of the monthly status review (see CDRL PM-1). The collected software measures will support the analysis of both software product quality and schedule/effort/cost performance. The collection and reporting of metrics will be automated to the fullest extent practical.

4.2.11 Operational Simulator Software

The Contractor **shall** develop and submit an Instrument Simulator Specification (CDRL SE-13), documenting the functional requirements for the Operational Simulator (OpSim). This document **shall** align with the relevant instrument simulator, requirements and interfaces specified in the Instrument Simulator Requirements Specification.

The Contractor **shall** develop and maintain Simulator Software detailed design; the Contractor **shall** also continue to maintain Simulator Software Development Folders (SDF) and Simulator SENs. The content and process to be used for the SDF and SEN **shall** be documented in the SDP. Delivery of the Simulator SDF and SEN to the Government is not required, but they **shall** be made available for audit by the Government at the Contractor's facility. Issues identified **shall** be addressed within 60 days.

4.2.11.1 Test and Verification

The Contractor **shall** plan and implement the testing of instrument-spacecraft interfaces with the Spacecraft-Instrument Interface Simulator (SIIS) provided by the spacecraft Contractor (TBR). Technical support will be provided by the spacecraft Contractor at the instrument simulator provider facility. This will support early risk reduction of instrument-spacecraft interfaces.

The instrument Contractor **shall** support the spacecraft activity to verify the performance of the flight software by providing flight quality procedures and scripts and providing technical support to the spacecraft Contractor performed testing including completing a Government approved Day-In-The-Life test which will be conducted at the Spacecraft simulator provider facility.

The Contractor **shall** deliver an Instrument Simulator Interface Verification Report (CDRL SE-14).

4.2.11.2 Simulator User's Guide

The Contractor **shall** update and deliver a Simulator User's Guide (CDRL SW-12) describing the installation, build, and operating procedures for the Simulator Software.

4.2.11.3 Simulator Training Materials

The Contractor **shall** update and deliver simulator training materials for the OpSim (CDRL OO-6).

4.2.12 NASA Independent Verification and Validation Support

The Contractor **shall** ensure that all software documentation and code required for NASA's Software IV&V effort is made available to NASA IV&V personnel if directed by the Government (TBR). This includes access to all software reviews and reports, developer plans and procedures, software code, software design documentation, and software problem reporting data. Wherever possible, the Contractor **shall** permit electronic access to the required information or furnish soft copies of requested information to NASA IV&V personnel.

The Contractor **shall** review and disposition all NEON Program-approved IV&V findings and recommendations. A Contractor point of contact **shall** be assigned and available to NASA IV&V personnel, as required, for questions, clarification, and status meetings.

4.2.13 Software Maintenance

The Contractor **shall** prepare and deliver an Instrument Software Maintenance Plan (CDRL SW-14) describing the plans, products, schedule, tasks, resources (software, hardware, facility) requirements, risks, and recommended contingencies for the maintenance of the flight software, instrument simulators, and ETU/GSE/STE systems.

The Contractor **shall** develop and maintain the Instrument flight software, simulators, and documentation, along with the environments, emulators, and test software necessary to develop and verify these systems through the duration of the Contract.

Any Contractor developed software development and/or validation tools (test bed) used by the Contractor to qualify and maintain the Instrument flight software **shall** be maintained by the Contractor for the duration of the Contract.

The Contractor **shall** maintain any special hardware (non-commercial such as command and data handling (C&DH) components and power supplies) for the duration of the Contract.

The delivery of a flight software patch or update **shall** be executed via the processes specified in the Project Flight Software and Table Update Delivery Process (TBS) (i.e., by sending a CD with the patch or uploading to a SharePoint folder).

4.2.14 Sensor Data Sets

The Contractor **shall** prepare and deliver sensor test data sets simulating 72-hour continuous science data (CDRL SW-15). The sensor test data sets will be determined by the Contractor and **shall** include two sets: one representing normal instrument operation and one with as many different configuration modes and data types as possible. This data will be used to support ground system development and end-to-end mission rehearsals.

4.2.15 Flight Software Deliverable

The Contractor **shall** deliver and periodically update a Flight Software Development Environment (CDRL SW-TBD):

- The Flight Software Development Environment (FSWDE) **shall** consist of the hardware and software systems used to develop and maintain the flight software, such as:

computer workstation, editors/integrated development environments (IDE), modeling tools, compilers, linkers, loaders, debuggers, and associated CM and user interface tools.

- The FSWDE **shall** contain all items needed to prepare the flight software for the target processor such as: flight software source code, real time operating system, board support package software, object-oriented models, databases, miscellaneous development tools, utilities, scripts, procedures, and instructions.
- The FSWDE **shall** include a library of hard- and/or soft-copies of all relevant documentation for its component hardware, commercial software and utilities, and custom software and utilities.
- The FSWDE **shall** produce flight software image files that could be loaded onto the Spacecraft and onto Simulators/Test Systems via the NOAA Enterprise Ground System, and that are bit-wise identical to those produced at the Contractor's facility.
- The FSWDE **shall** mirror to the extent practical the hardware and software configuration being used for flight software development at the Contractor's facility.
- The FSWDE **shall** be delivered to the Government's facility within 30 days of flight software Critical Design Review (or at equivalent flight software maturity).
- After initial delivery, the FSWDE **shall** be updated with any added or modified elements quarterly and within one week of any flight software release/delivery.
- The FSWDE **shall** be initially populated with any flight software being reused or inherited from an earlier mission as a baseline starting point.
- Where applicable, COTS software **shall** be delivered with the initial licensing period paid by the Contractor.
- Third party software **shall** be provided to the Government.

4.3 Hardware

The Contractor **shall** deliver hardware as defined in the Contract.

4.3.1 Engineering Test Unit

The NASA Systems Engineering Handbook defines an Engineering Unit as: “A high-fidelity unit that demonstrates critical aspects of the engineering processes involved in the development of the operational unit. ETUs are intended to closely resemble the final product (hardware/software) to the maximum extent possible and are built and tested so as to establish confidence that the design will function in the expected environments. In some cases, the engineering unit will become the final product, assuming proper traceability has been exercised over the components and hardware handling.” A high-fidelity ETU is a critical tool for the program to manage technical and programmatic risk.

The Contractor **shall** conduct an Engineering Development Program to raise confidence in the instrument design and to minimize risk, cost, and schedule for the FM development, this will include the EPRs called out in Section 2.4, Design Engineering. At a minimum, the Engineering Development Program **shall** include development of a stand-alone ETU that is representative of the form, fit and function of the FMs.

The ETU should be used to:

- Evaluate the flight hardware design.
- Evaluate FM test program.

- Dry-run the assembly build process, develop assembly and test procedures, initial run of test procedures, and test scripts, and identify any necessary mechanical GSE.
- Validate electrical and data interfaces design.
- Long term use for reach back and reach forward investigative support for anomalies, and updates for the flight instrument series.
- Identify and incorporate design updates for test procedure, execution improvement updates, and GSE updates.

The ETU **shall** be vacuum compatible, mechanically equivalent, and electrically equivalent, and use the same fabrication methods as the FMs to the extent possible. The ETU may be used for the development of Recommended Operating Procedures (ROP), validation of GSE, etc. Non-flight parts may be used provided form, fit, and functionality is maintained. Decisions regarding non-flight part substitution **shall** be made by the PCB and the MRB.

The Contractor **shall** prepare and deliver an ETU Development and Test Plan in accordance with CDRL ETU-1 and submit an ETU Requirements Alignment Matrix that documents differences between the ETU and the flight instrument, in accordance with CDRL ETU-2. These CDRLs will serve as the backbones for the ETU Pre-Build Review and ETU Test Summary Review.

The build, assembly, and functional test of the ETU **shall** be completed prior to the Instrument-level CDR. If there is a conflict between the ETU testing and the CDR that will drive the critical path schedule, the Contractor **shall** notify the Government and attain Government approval if the CDR is held prior to the ETU testing completion and ETU Test Summary.

The Contractor **shall** retain and maintain the ETU at the Instrument Contractor's facility during the FM development, test, and operations for the duration of the program.

4.3.2 Flight Instruments

The Contractor **shall** utilize the serial number to identify each instrument. The flight units **shall** be built from formally released drawings, utilizing Contractor's approved parts, materials, processes, and procedures. The flight units **shall** be successfully tested by the Contractor in accordance with the Specification and the approved test procedures/levels/durations (CDRL IT-3). Test procedures may be tailored as necessary to achieve the equivalent of acceptance as defined in the ICD and PRD. Any proposed tailoring **shall** be in accordance with CDRL IT-1.

The Contractor **shall** provide the survival heaters that will be powered by the spacecraft.

4.3.2.1 First Flight Model

The Contractor **shall** develop and deliver the first SMBA FM, FM1. FM1 **shall** be tested to protoflight levels as per Section 7 of the PRD.

The FM1 effort **shall** include fabrication, assembly, test, maintenance, storage, delivery, post-delivery, post-launch, and on-orbit operations support per the Contract.

4.3.2.2 Additional Flight Models

The Contractor **shall** develop and deliver additional FMs as per the Contract.

The additional FMs **shall** be tested to acceptance levels as per Section 7 of the PRD. If modifications to the design are made between FMs, the Contractor and the Government will come to agreement on the test levels.

Note: For thermal vacuum verification, all hardware, including follow-on units, should be tested to protoflight levels per the PRD.

The effort **shall** include fabrication, assembly, test, maintenance, storage, delivery and post-delivery, post-launch, and on-orbit operations support as per the Contract.

4.3.2.3 Spares

The Contractor **shall** procure any necessary additional spares to ensure continued compliance with the requirements in Section 4.1.

The Spares Plan **shall** be updated in accordance with CDRL SE-12.

4.3.3 Ground Support Equipment

The Contractor **shall** develop and maintain GSE, including the STE, in accordance with the Contract. The GSE **shall** include specialized equipment necessary to support accommodate the instruments at all facilities during I&T, transportation, storage, satellite I&T, launch site, and any other location. This should include GSE spares.

The Contractor **shall** provide the following information regarding GSE:

- a) Number and dimensions of racks.
- b) Location of racks.
- c) Phase of I&T the test racks are required.
- d) Power requirements.

The Contractor-provided Mechanical/Electrical GSE **shall** include all items necessary for testing, deploying, stowing, and protecting the instrument, including, but not limited to:

- STE – Interfaces to the SMBA instrument to:
 - Provide power.
 - Monitor health & Status.
 - Record and store data/files.
 - Simulate interfaces to the spacecraft (e.g., C&DH, etc.).
 - Control and monitor calibration targets.
 - Instrument commanding.
 - Provide timing.
- Hard Protective Cover for antenna apertures.
- Mechanical GSE for use during Vibration testing and Thermal Balance/Thermal Vacuum testing.
- Drill template.
- Electrostatic discharge bags.
- EMI cover.
- Calibration System for Ambient Testing.

-
- Calibration Targets (for Thermal Vacuum (TVAC) Instrument Level Testing).
 - RF blocker for EMI Testing.
 - S/C Emulator (delivered from the S/C provider).
 - Flight software and Firmware Test Bed.
 - Spacecraft test and operations language GSE.
 - RFI generator (TBR).

External GSE Calibration Targets configuration **shall** be capable of accommodating Government provided targets in place of Contractor targets.

The Contractor **shall** provide two (TBR) additional sets of thermal control equipment for exclusive use at the spacecraft integration facility(ies).

4.3.4 Operational Simulator

The Contractor **shall** develop, test, and deliver three (3) OpSims.

The SMBA instrument OpSim is used by the NEON Series-1 team to prepare the launch readiness activities, verify satellite operational loads, flight software patches, maneuver planning, ground system development, verification and validation and maintenance of operational procedures and testing, flight product validation, test/operations procedure development and validation, and operator training.

To assist in the development, testing, and operational evaluation of flight software patches, the Spacecraft Simulator **shall** run the same flight software image(s) as the actual flight system.

The OpSim **shall** operate in real-time.

NOTE: In this context, “real-time” means that one unit of wall-clock time equals one unit of simulation time, and that all interface timing requirements for the actual flight software executing in the environment/context of its host flight hardware platform will be simulated in like manner.

4.3.5 Software Testbench

The Contractor **shall** develop and test a SMBA Software Testbench with the following functionality:

- a) Command and data handling.
- b) Scheduling of events.
- c) Interrupt handling.
- d) Control of interfaces.
- e) RFI detection algorithms.

(The Software Testbench **shall** be developed, tested, and documented in accordance with CDRLs SW-4 Software Requirements Specification, SW-6 Software Test Plan, SW-18 Software Test Reports, SW-9 Software Version Description Document, and SW-11 Software Acceptance Review Data Package.)

4.4 Deliverable Documentation

Deliverable documentation is specified in the CDRL. All data **shall** be delivered to the Government as specified in the Contract. Electronic submittal of CDRL items is required. Delivery dates shown in the CDRL are the dates the documents are to be received at the Government's facility.

Each data item **shall** be prepared in accordance with the requirements defined in this SOW and accompanying CDRL. The Contractor may supply existing documentation if the existing documentation encompasses the requirements defined in this SOW. All CDRL items **shall** be delivered for Approval, Review, Information, or as Previously Delivered Acceptance. See the CDRL for the definition of each category.

For all document submissions, the Contractor **shall** send an email with a formal contracts letter to the Government informing the Government of the virtual delivery of the document.

CDRL submissions **shall** be complete, unabridged, and unredacted. To the extent that Proprietary Data is employed to support program execution, such data **shall** be appropriately marked and included.

The Contractor **shall** be responsible for the packaging and delivery of documents and data in the CDRL.

All CDRL submittals **shall** be protected by encryption or through the use of other secure means approved by the Government against unauthorized disclosure including interception and decryption by foreign nationals. Prior to delivery, the Contractor **shall** implement all safeguards necessary to ensure data is protected in accordance with the Export Administration Regulations except when expressly authorized by a Technical Assistance Agreement.

5 TEST PROGRAM

The Contractor **shall** prepare and conduct a test program that meets qualification, protoflight, and acceptance levels using GSFC-STD-7000 General Environmental Verification Standard (GEVS) as a guideline and as specified in the SMBA PRD. The test program **shall** demonstrate that all applicable specified requirements are satisfied. The test program **shall** include tests as necessary at all levels of assembly. The hardware test program **shall** cover all phases of the development and manufacture in accordance with the requirements identified in this SOW and the PRD. The Contractor **shall** provide a plan that includes any recommended tailoring for the instrument environmental tests for Government approval in accordance with CDRL item IT-1.

All composite structures and structural bonded joints **shall** be proof tested, regardless of safety factor.

Cabling electrically and physically equivalent to flight cables **shall** be used for all EMC testing. A Government agreed-to design, constructed with expected flight wire groupings, twisting, shielding, and shield terminations. The Contractor **shall** conduct a test program to demonstrate that all applicable specified requirements are satisfied. The test program **shall** include tests as necessary at all levels of assembly.

The Contractor **shall** provide access to facilities and data to support independent review of test data by the Government team during instrument performance and environmental tests. Tools utilized by the Contractor to verify instrument performance **shall** be made available to the Government.

Contractor **shall** take and maintain test data in accordance with the Sensor Performance Specification, the approved test plan (CDRL IT-1) and the requirements. Problems, anomalies, failures, etc., **shall** be reported and treated in accordance with the IMAR document, as tailored by SMBA Mission Assurance Compliance Matrix (**TBS**) within Contract **TBS**, Clause **TBS**, Attachment **TBS**, identified in Section IV for investigation, resolution, and corrective action.

Items that incorporate changes in design, manufacturing processing, environmental levels, or performance requirements **shall** be re-qualified.

Qualification by similarity **shall** be permitted only with the concurrence of the Government.

The following is a sample SMBA Instrument Test Flow:

- Spectral Response Function (SRF) Test
- Antenna Pattern Measurement Test
- First Comprehensive Performance Test (CPT)
- Antenna Alignment Test
- Vibration Tests
- Antenna Alignment Test
- Momentum Compensation Test
- Weight and Center of Gravity Test
- EMI Tests

- RFI Detection Algorithm Testing
- Thermal Vacuum Cycling Test
- Thermal Vacuum Balance Test
- Thermal Vacuum Calibration Tests
- Antenna Alignment Test
- Final CPT

SRF and antenna pattern measurements **shall** be measured as defined below. All other tests in the Suggested Test Flow **shall** be done in accordance with the requirements specified in the SMBA PRD.

5.1 Spectral Response Functions

The SRF **shall** be measured and fully documented for each band/channel of each instrument in accordance with PRD requirements in PRD Section 4.10 and CDRL CV-6.

SRF data **shall** be characterized at a minimum of three temperature conditions: coldest predicted operational temperature, warmest predicted operational temperature, and the median temperature between the coldest and warmest predicted operational temperatures.

SRF data **shall** accommodate SRF changes over the operational lifetime of the instrument.

The gain range of SRF data **shall** be at minimum the achieved channel rejection magnitude per the PRD and measure the noise floor. SRFs need to accurately represent the noise floor.

5.2 Radio Frequency Interference

The Contractor **shall** test the SMBA instrument's capability to detect man-made RFI in the bands identified in PRD Table 4.2.1.

The Contractor will decide the best means of doing this (e.g., data sets, data ports, RFI generator, RFI Test Bench, etc.).

RFI testing **shall** include an end-to-end test.

The Contractor **shall** develop and deliver RFI test cases and perform RFI detection analysis (CDRL SE-7j) for what is currently seen from both terrestrial and space-based sources and from what can be reasonably projected in the future.

The algorithms utilized in the detection schemes **shall** be chosen by the Contractor and approved by the Government.

5.3 Antenna Pattern Measurements

The antenna radiation characteristics of each instrument ETU and FM **shall** be measured and documented by the Contractor. For each SMBA channel, measurements **shall** include beam pointing, beamwidth, beamwidth variation, and beam efficiency.

The Contractor **shall** permanently affix an optical alignment cube that **shall** be used to reference antenna pattern measurements and provide the Spacecraft integrator a known relative indicator of boresight alignment in accordance with PRD Section 5.7.1.

For SMBA analog frequency bands, antenna patterns **shall** be measured at the center frequency of the corresponding frequency band.

For SMBA hyperspectral frequency bands, antenna patterns **shall** cover the frequency bands as defined in the PRD in Section 4.1. At a minimum, antenna patterns **shall** be measured at the two band edge frequencies and at the band center frequency. For V-Band, an additional two frequencies (one on each side midway between the band edge and the band center) **shall** be measured in accordance with CDRL SE-7j.

5.3.1 Plane Pattern Cuts

At each frequency, a minimum of three cuts **shall** be made to obtain radiation patterns.

These cuts **shall** be the principal plane of the antenna (plane of symmetry of the antenna, also referred to as the down track plane), a plane 90 degrees from the principal plane (also referred to as the cross track plane), and a 45-degree diagonal cut.

5.3.2 Measurement Polarization

For all channels, each cut **shall** be measured at two polarizations, Co-polarized (0°) and Cross-polarized (90°). Note: The 0° is parallel to the principal direction of polarization at the electrical boresight (beam center) of the beam position in question.

5.3.3 Dynamic Range

The minimum dynamic range of the antenna pattern measurements **shall** be 70 dB.

5.3.4 Amplitude Accuracy

The amplitude accuracy of the antenna pattern measurements **shall** be less than $\pm 3\%$ in dB values throughout the dynamic range.

5.3.5 Angular Range and Interval

Each pattern **shall** cover a total angular range of 360° , i.e., plus and minus 180° from electrical boresight. The angular intervals **shall** be a maximum of one-fifteenth ($1/15^{\text{th}}$) of the antenna Half Power Beamwidth.

5.3.6 Beam Positions Required

The antenna pattern measurements **shall** be made at beam positions corresponding with:

- Beginning, middle, and end of the Earth-view sector.
- Cold calibration reference beam position.
- Warm calibration reference beam position.

5.4 Test Readiness Reviews

TRRs for the instrument are defined in 1.5.1.10.

TRRs for instrument subsystems **shall** be performed as part of the applicable subsystem peer reviews, as defined in the Engineering Peer Review Plan (CDRL PM-3).

5.4.1 Test Process Decision Meetings

The Contractor **shall** document Consent to Proceed (CTP), Consent to Break (CTB), Consent to Integrate (CTI), or any other related test process decision meetings and the outcomes and make available upon request any data presented or referenced in support of any decision related to the forward processing of the Instrument. The Contractor **shall** invite the Government to attend these meetings/reviews. The Contractor **shall** specify in all in-process and end-item test procedures where Government approved CTP, CTB and/or CTI is anticipated (RE-8).

5.5 Pre-Environmental Review

The Contractor **shall** present the description and results of the Instrument Pre-Environmental Test program and demonstrate their readiness for environmental testing. The PER Data Package (CDRL RE-4) **shall** contain all relevant instrument information required to satisfy paragraph 8.2 (Criteria for a Successful System Integration Review) of GSFC-STD-1001, Criteria for Flight and Flight Support Systems Life Cycle Reviews.

5.6 Instrument Test and Calibration

5.6.1 Instrument Program Test Plan

The Contractor **shall** prepare and deliver a test plan describing the methods, policies, organization, and flight unit test program in accordance with CDRL IT-1. The Contractor may submit separate plans for each phase of the test program or one plan covering all phases.

5.6.2 Instrument Calibration

The Contractor **shall** perform through the antenna thermal vacuum calibration of the SMBA ETU and all FMs.

The system calibration test program **shall** produce the information required to evaluate the orbital measurement with a noise equivalent differential temperature (NedT) precision better than the required values listed in the PRD.

The system thermal vacuum calibration test program **shall** emulate “test-as-you-fly” conditions.

The Contractor **shall** collect calibration data in operational mode.

The calibration environment for the instrument **shall** emulate the mission thermal radiation environment.

The Contractor **shall** provide a variable Earth scene target with a radiometric brightness temperature that varies from 80K to 330K.

The Earth Scene Target temperature **shall** be capable of being set to within ± 1 K of its pre-selected value. The target temperature is determined from the weighted mean of temperature sensors imbedded in it.

The Contractor **shall** provide a cold space calibration target representative with a radiometric brightness temperature as close to the on-orbit cold space brightness temperature as possible. No warmer than 80K is desired.

The target temperatures are determined from the weighted mean of temperature sensors imbedded in them. During the period that calibration data is taken, the indicated temperature change of each sensor taken individually **shall** be less than 0.1 K. If any sensor indicates a target temperature change in excess of 0.1 K, all data for that run will be discarded.

The Contractor **shall** perform calibration at a minimum of three instrument temperatures representing the SMBA instrument low operational, mid operational, and high operational temperatures (TBS). Actual instrument operational temperatures and control temperature locations will be agreed upon between the Contractor and the Government.

The instrument temperature **shall** be set to within ± 1 K of its preselected value. The temperature will be stabilized during the data run for any given target temperature. If the interface temperature varies by more than ± 1 K, all data for that run will be discarded.

The instrument internal temperature **shall** be taken at one point within the instrument where there is a flight temperature sensor. The temperature **shall** be set within ± 1 K and stabilized sufficiently during the data run for any given target temperature. If the instrument internal temperature varies by more than ± 0.5 K, all data for that run will be discarded.

The Contractor **shall** perform hysteresis measurements at a minimum of two variable Earth scene target temperatures at each of the instrument operational temperatures. Actual target temperatures will be agreed upon between the Contractor and the Government.

The Contractor **shall** prepare and deliver a Calibration Plan that includes all aspects of calibration both ground and on-orbit. This will include intercomparison and intercalibration as part of a Calibration Plan per CDRL CV-1. This may include a Government-furnished equipment calibration target. Intercalibration may be done at GSFC (TBR).

Procedures for Thermal Vacuum Calibration/Validation (Cal/Val) **shall** be provided in accordance with CDRL CV-2. Procedures for Cal/Val and intercalibration **shall** be provided in accordance with CDRL CV-2.

Cal/Val reports and summaries for the ETU and each FM **shall** be provided in accordance with CDRL CV-3.

A Radiometric Math Model for the SMBA Instrument **shall** be developed and provided in accordance with CDRL CV-4.

A Calibration Data Book for the ETU and each FM **shall** be provided in accordance with CDRL CV-5.

Pre-launch calibration parameters required for data processing **shall** be delivered in accordance with CDRL CV-5.

Calibration, intercalibration, and data processing algorithms **shall** be provided in accordance with CDRL CV-7.

The Contractor **shall** perform Point & Stare operations as defined in PRD Section 4.9.3, Scan Characteristics, and Section 5.6.2.4.2, Instrument Diagnostic Mode, at a minimum of two variable Earth scene target temperatures at each of the tested instrument temperatures to determine the Noise Power Spectral Density (NPSD) curves like that shown in Figure 5.1.2-1 for each SMBA channel as defined in PRD Table 4.1.2. The NPSD is an important characteristic of all radiometer channels relating quantities like NEdT, $1/f$ noise, gain variations, and striping. Actual target temperatures will be agreed upon between the Contractor and the Government.

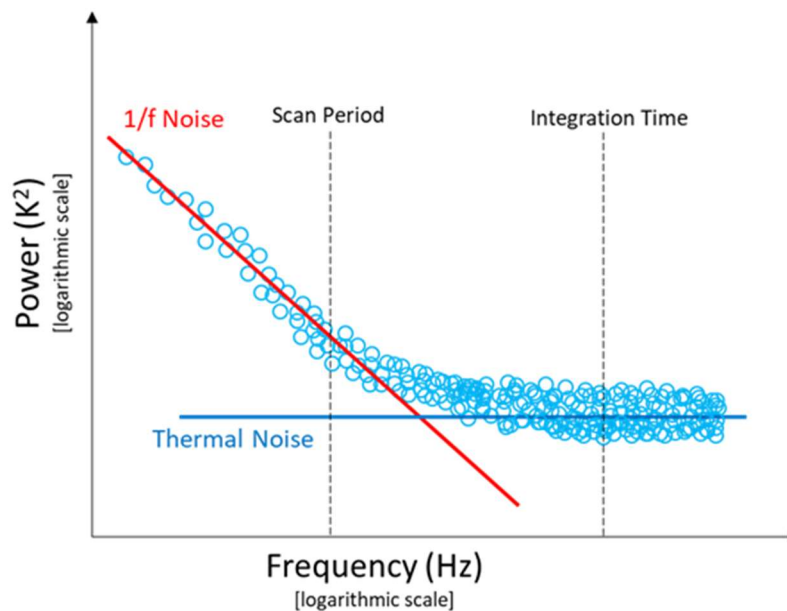


Figure 5-1. Notional Noise Power Spectral Density

The Contractor **shall** prepare and deliver an Intercalibration Plan in accordance with CDRL CV-1 describing tests to be conducted for calibrating the radiometric response of the instruments against each other and a ground-based standard.

The Contractor **shall** prepare and deliver the SRF data for the SMBA Instrument in accordance with CDRL CV-7.

5.6.3 Calibration Test Equipment

The Contractor **shall** provide one set of Calibration Test Equipment (CTE).

The CTE **shall** be designed and built to be used in a thermal vacuum environment.

The CTE **shall** include calibration reference targets representing the on-orbit cold space view scene and on-orbit Earth viewing scene.

All targets **shall** have a target emissivity of no less than 0.9999 for all SMBA channels.

All targets **shall** be large enough to fill the instrument field of view when the instrument is pointed toward the target.

The CTE cold space view calibration reference target(s) **shall**:

- Contain 4-wire platinum resistance thermometers (PRT) imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target.

The CTE Earth viewing scene calibration reference target(s) **shall**:

- Produce a minimum of 11 radiometric brightness scene temperatures corresponding to target physical temperatures from 80K to 330K in 25K steps.
- Contain a number of 4-wire PRTs imbedded and located in the target sufficient to measure the physical temperature of the target and measure the vertical and horizontal temperature gradients within the target.

5.6.4 Test Software Validation Plan

If automated or semi-automated testing is planned or used, the Contractor **shall** carry out a software validation process as documented in the Instrument SDP (SOW Section 4.3.1).

5.6.5 Test Procedures

The Contractor **shall** prepare Detailed Test Procedures for all hardware to be delivered (CDRL IT-3). For those supporting documents that contain items that are unique to specific units (i.e., test data and configuration information), representative documents **shall** be submitted.

The Contractor **shall** deliver as-run, instrument test procedures in accordance with CDRL IT-3. The Government will be responsible for any translation of test procedures to ensure compatibility with Government hardware or software.

5.6.5.1 Notification of Tests

The Contractor **shall** notify the Government no later than five days in advance of instrument and lower-level acceptance tests. Tests will not be delayed if Government's representative is not available for any test unless Government presence is required per the Contract, i.e., mission assurance requirements.

5.6.6 Test Reports and Test Data

Test reports **shall** be submitted after completion of each test in accordance with CDRL IT-2.

5.7 Verification and Validation

5.7.1 Verification & Validation Plan

The Contractor **shall** develop and deliver a comprehensive Verification and Validation (V&V) Plan and associated Requirements Verification Matrix in accordance with CDRL SE-4.

5.7.2 Verification Reports

The Contractor **shall** provide a verification report in accordance with CDRL SE-5.

6 DELIVERY

The Contractor **shall** be fully responsible and liable for the preservation, packaging, product marking, packing and delivery of all hardware and documentation in accordance with this SOW and all applicable documents of Sections VI and VII to assure safe arrival at destination.

6.1 Acceptance or End Item Data Package

An acceptance or EIDP **shall** be prepared for each flight deliverable instrument and the ETU. The requirements for the EIDP are specified in the CDRL MA-32 data item description.

6.2 Packaging, Handling, Storage, and Transportation

The Contractor **shall** provide Packaging, Handling, Storage, and Transportation (PHS&T) Plans and Procedures in accordance with CDRL IT-4. The Contractor **shall** provide the personnel, facilities, and hardware necessary to prepare and pack the instrument and its GSE for shipment and **shall** be responsible for the transportation and shipment of the material to the designated spacecraft facility, or other required location. The Contractor **shall** prepare, pack, and ship all instrument subsystems, systems, and the simulators between the places of manufacture, and integration and test, and storage and delivery in appropriate shipment containers. The Contractor **shall** prepare, pack, and ship all related calibration, mechanical, and electrical GSE required to support any instrument subsystems, systems, and simulators during each phase of test, integration, and launch preparation.

The PHS&T (IT-4) **shall** include long term (10 years) storage plans for the ETU, FM1, and additional FMs as required.

The Contractor **shall** design, develop, and deliver shipping and storage containers necessary for the ETU and additional FMs as needed based on the IMS and the Contractor's Integration and Test Plan.

The instrument shipping container **shall** accommodate the instrument(s), as well as any shipping environment-related monitoring equipment and contamination witness plates.

Instrument EGSE **shall** make accessible instrument-relevant test data, such as chamber telemetry, analog test point, accelerometer, and thermocouple data. The content of this data will be determined jointly by the instrument and spacecraft providers and defined in the applicable instrument ICD.

The electrical interface between the spacecraft bus and the instrument **shall** be checked out prior to instrument delivery to the spacecraft using an electrical interface simulator provided by the spacecraft Contractor.

6.3 Pre-Ship Review

Prior to shipment, the Contractor **shall** conduct a PSR in accordance with CDRL RE-5 or delta review. All documentation required to manufacture, test, calibrate, and deliver the hardware as well as all test data of the specific hardware under review **shall** be available for the Government's review at the PSRs.

6.4 Delivery, Checkout, and Acceptance

The Contractor **shall** unpack the flight hardware and GSE upon its arrival at the spacecraft Contractor's facility, perform a visual inspection to detect physical damage, set up a bench test environment, and perform a post-delivery functional check out of the instrument.

The spacecraft Contractor will provide facilities, security, storage, standard cleaning supplies, and standard test equipment (multi-meter, probes, grounding, ionized air, etc.). The Contractor **shall** provide a bench check-out test report including comparison to the identical pre-ship test performed at the Contractor's facility. The Contractor **shall** compare the instrument post-shipment functional test results with the pre-shipment functional test results and provide comparative analyses and results in accordance with CDRL IT-7.

Government conditional acceptance will occur after acceptance testing confirms successful completion of mechanical and electrical integration of the instrument with the spacecraft. The Contractor **shall** be responsible for the instrument handling and operation from delivery until lifted "on-hook" by the spacecraft Contractor for mechanical integration with the spacecraft. The spacecraft Contractor will be responsible for maintaining and controlling a safe environment for all instrument hardware and personnel from the time of arrival at the spacecraft Contractor's facility until delivered to the launch site, or ready to ship for return to the Instrument Contractor, as directed by the Government.

6.4.1 Hardware Delivery Documentation

6.4.1.1 Operations Support Manuals

The Contractor **shall** provide the following Operations Support Manuals to maintain and monitor the instrument while in storage, and to operate it during satellite integration and test and on-orbit:

- a) The Instrument On-Orbit Commissioning Plan, CDRL OO-1.
- b) The Instrument User's Manual, CDRL OO-2.
- c) The Instrument Command and Telemetry Handbook, CDRL OO-4.
- d) Instrument On-Orbit Launch and On-Orbit Operations and Contingency Procedures, CDRL OO-5.

6.4.1.2 Technical Manuals

The Contractor **shall** provide the following documents with the auxiliary information needed for the operation of the instrument:

- a) The Software User's Guide, CDRL SW-12.
- b) The Simulator Description and Training Manual, CDRL OO-6.

6.4.2 Documentation Delivered with Hardware

Handling instructions/procedures and accompanying documentation may be inside the outer shipment package but **shall** be outside the sealed inner package and be accessible upon opening of the outer box, without having to remove the inner box.

6.4.3 Photographic Documentation

The Contractor **shall** produce, archive, and make available to the Government upon request, closeout photographic documentation of all assemblies during the manufacturing process and of the final integrated configuration "as flown."

6.4.4 Test Data Files

The Contractor **shall** provide interim data files of the test data taken during the instrument test sequence in the Contractor's format and with appropriate logs in a timely manner for real-time discussion of test results with the Government upon request. The Government will provide access to an approved electronic document repository (e.g., the external SharePoint where test data will be posted).

6.5 Sensor Database Support

The Contractor **shall** supply XML (TBR) formatted database inputs as required by the ground system.

6.5.1 Sensor Command and Telemetry Database Support

The Contractor **shall** deliver the Instrument Command and Telemetry Database (CTDB) (CDRL OO-8). The Contractor **shall** place the CTDB under formal CM. In addition, the Contractor **shall** prepare and deliver a CTDB Validation Plan (CDRL OO-10) that describes the process used to verify and validate:

- a. The instrument-level database (CDRL OO-8).
- b. The system-level database.

6.5.2 Instrument Mission Data Structure Definition Support

The Contractor **shall** deliver the Instrument Mission Data Structure Definitions (CDRL OO-9) to be used within the Missions Operation Element to identify and process mission data produced by the sensor.

6.5.3 Instrument Memory Table Support

The Contractor **shall** deliver the Instrument Memory Table Structure Definitions and baseline content for the loadable and dumpable Instrument Memory Tables, to be used to populate the Ground Reference Image, in accordance with CDRL OO-11.

6.6 Instrument Post-Delivery Support

This section describes the Contractor's staffing and material responsibilities to support the mission system implementation, which includes satellite integration and test, training, mission readiness support, launch site support, and support through and beyond operations hand-over through the life of the Contract. The Contractor **shall** be responsible for the operations support and performance evaluation of each instrument through satellite decommissioning.

The post-delivery period (from delivery of the SMBA through on-orbit support) is divided into the following sub-periods for the purposes of explicitly defining Contractor support:

- Post Delivery Support Planning (PDSP) = Activities prior to delivery.
- Post Delivery Support (PDS) = The period from delivery to Launch.
- Commissioning Support (CS) = The period from Launch to Launch + 6 months.
- Post Launch Support = The period beyond Launch + 6 months, through the duration of the Contract.
- Sustaining Engineering and Cal/Val Support.

The instrument Contractor **shall** provide satellite integration and test, mission operations readiness support and training, launch site support, and support through operations hand-over and the following CDRLs:

- OO-5 Instrument Launch and On-Orbit Operations and Contingency Procedures.
- OO-7 Constraints, Restrictions, and Warnings/Alerts Document.

The instrument Contractor **shall** provide the following inputs to the spacecraft Contractor for Instrument GSE and Flight Hardware testing as necessary as part of risk reduction testing and/or as part of satellite acceptance testing:

- Applicable instrument procedures or inputs to spacecraft procedures that are necessary to run the GSE, ETU, or Flight Hardware.
 - The inputs to spacecraft procedures capturing all activities, equipment, personnel, and support required for the Contractor's instrument to be ready for flight after being delivered to the spacecraft Contractor **shall** be documented through a PDS document in accordance with IT-7.
- Applicable instrument scripts/SW or inputs to spacecraft scripts/SW that are necessary to run the GSE, ETU, or Flight Hardware.

The Government's intent is to do this in a collaborative environment between the instrument Contractor, the spacecraft Contractor, and the Government.

6.6.1 General Roles and Responsibilities (PDSP and PDS Phases)

Prior to launch, the Contractor **shall** be the sole responsible party for mechanical instrument support, including, but not limited to, red and green tag removal and installation, cleaning, inspection, etc. The Contractor **shall** be responsible for the operation and physical handling of the instrument until the spacecraft Contractor accepts it for mechanical integration with the spacecraft, and whenever the instrument is de-integrated from the spacecraft. The Contractor **shall** be responsible for operating and maintaining their GSE at the spacecraft Contractor and launch site facilities. The Contractor **shall** be responsible for shipping the instrument GSE from the launch site payload processing facility in a timely manner after launch. The Contractor **shall** be responsible for monitoring and reporting the instrument health and safety at all times the instrument is powered. The Contractor **shall** have personnel on-site during powered operations.

6.6.2 System Integration and Test Support

6.6.2.1 Instrument Contractor Personnel and Facility Support

The Contractor **shall** provide personnel necessary to support integration and test activities at both the Contractor's facility and the spacecraft integration facility, including: performing instrument bench checkout; providing technical support during integration of the instrument to the spacecraft; performing all required repairs and re-tests; supporting limited and comprehensive performance testing throughout satellite qualification; performing thermal, mechanical, and electrical close-out; cleaning and monitoring contamination as needed; and supporting electrical system checkout following integration. The Contractor **shall** provide the survival heaters that will be powered by the spacecraft.

To prepare for the activities listed above, the Contractor **shall** be required to support periodic virtual and in person face-to-face TIMs with NASA and the spacecraft Contractor. These TIMs

will be used to define satellite Limited Performance Tests, satellite CPTs, satellite environmental tests, and miscellaneous test details.

The Contractor **shall** support TVAC Dry Run activities to ensure compatibility of the instrument GSE with spacecraft vendor facility as well as to train/familiarize the SMBA team with the facility prior to flight instrument delivery.

6.6.2.2 Instrument Performance Evaluation

The Contractor **shall** support instrument performance evaluation (e.g., data reduction and trending) during all phases of satellite integration and testing, including, test support, anomaly resolution support, and instrument data archival at the Contractor's facility, with electronic access provided to the Government, as required. Data resulting from these evaluations and tools utilized by the Contractor **shall** be made available upon request by the Government.

6.6.2.3 Instrument Data Products Support

The instrument Contractor **shall** provide the necessary instrument operating and test procedures (CDRL IT-3), parameters, screen displays, databases (CDRL OO-11), red and yellow limits definition (CDRLs OO-4 and OO-8), and instrument constraints (CDRL OO-7). The instrument Contractor **shall** provide support for the conversion and validation of these data. This support **shall** include as a minimum a review of the converted documentation (page display printouts, procedure printouts) as well as validation in a satellite test environment. Additionally, the instrument Contractor **shall** provide instrument safety information and may be required to support periodic safety meetings. Instrument Data Sets for use in Ground Segment testing **shall** be provided in accordance with CDRL 3.

The Contractor **shall** prepare and accommodate remote network access from spacecraft Contractor facility to the instrument vendor facility in order to transmit test data efficiently to the instrument vendor facility for off-line analysis.

6.6.2.4 Interface Testing

The Contractor **shall** support interface testing between their instrument (or a reasonable facsimile) and the spacecraft C&DH system (or a reasonable facsimile), including interfacing and testing the instrument (simulators, Engineering Development Unit, Flight Module, or other) and instrument GSE with the spacecraft data systems and C&DH simulators (including, but not limited to, the SIIS) to demonstrate proper operation. In support of these activities, the Contractor **shall** maintain, operate, and deliver instrument GSE hardware and software to the interface testing facilities.

As part of the above defined interface testing, the Contractor **shall** plan and implement support of spacecraft to payload interface electronics development risk mitigation testing. Technical support will be provided by the Contractor at the spacecraft Contractor facility. This will support early risk reduction of instrument-spacecraft interfaces.

As part of the above defined interface testing, the Contractor (in conjunction with the spacecraft Contractor) **shall** plan and implement the testing of instrument-spacecraft interfaces with the SIIS provided by the spacecraft Contractor. Technical support will be provided by the spacecraft Contractor at the instrument simulator provider facility and any training necessary to facilitate

such support by the spacecraft Contractor will be provided by the instrument simulator provider. This will support early risk reduction of instrument-spacecraft interfaces.

6.6.2.5 Engineering Test Unit Support

The goal of the ETU phase of satellite I&T is to perform instrument mechanical and electrical integration; instrument power, command, telemetry, and data interface verification; and procedure and database validation. This may include thermal testing at ambient pressure and/or vacuum and EMI testing. The Contractor **shall** support ETU efforts related to risk reduction for flight unit integration and test.

Additional risk reduction activities utilizing the SMBA ETU at the spacecraft Contractor facility include, but are not limited to:

- SIIS testing with SMBA ETU.
- Inter-Operability Testing with a representative SMBA unit.
- EMI risk reduction testing with SMBA ETU.

6.6.2.6 Bench Checkout

The Contractor **shall** provide technical support for the instrument at the spacecraft facility. The goal of the bench checkout at the spacecraft facility is to certify that the instrument has survived shipment to the spacecraft facility by replicating the pre-ship limited functional test. The Contractor **shall** define a limited functional test for the instrument, using the instrument GSE, for use after delivery to the spacecraft Contractor's facility. The Contractor **shall** be responsible for performing this test prior to integrating the instrument to spacecraft. The instrument EGSE **shall** be used to perform the stand-alone Bench Acceptance Test when the instrument arrives at the spacecraft Contractor.

6.6.2.7 Flight Model Support

The Contractor **shall** provide a FM instrument, with necessary GSE and personnel, to support integration and testing with the spacecraft. Satellite integration and test will encompass mechanical integration (including alignment), electrical integration, limited and comprehensive performance testing at ambient, satellite environmental qualification, and post-environmental functional and performance testing. The Contractor **shall** support the satellite Pre-Environmental Review held prior to satellite environmental testing, and the satellite Pre-Ship Review held prior to shipment to the launch site.

6.6.3 Flight Vehicle Test Suite Support

The Contractor **shall** provide post-delivery support for the delivered simulators.

6.6.4 Launch Site Support

The Contractor **shall** provide launch site support at the payload processing facility as required. This support **shall** include the following:

- Limited satellite functional testing prior to integration to the launch vehicle.
- Final thermal, electrical and mechanical closeout of the instrument.
- Final red tag/green tag removal and installation.
- Final instrument inspection and cleaning.
- Anomaly support.

The Contractor **shall** support Flight Readiness Review, Launch Readiness Review, and other reviews as required.

6.6.5 Mission System Integration Support

The Contractor **shall** support planning and execution of mission systems integration activities through satellite operational hand-over, including training and mission readiness support as described below.

6.6.5.1 Integrated Mission Timeline

The Contractor **shall** support development of Integrated Mission Timeline (IMT), including:

- a) Support for regular IMT Working Group.
- b) Review IMT prior to major stakeholder reviews and attend reviews.

6.6.5.2 Operations Reviews

The Contractor **shall** support operations reviews including but not limited to MOR, ORR, Mission Readiness Review, and Operations Handover Review, specifically:

- a) Review instrument-specific information in review materials for accuracy and completeness.
- b) Provide Q&A support during instrument-specific sections of review presentations.

6.6.5.3 Rehearsal Anomaly Definition

The Contractor **shall** support Rehearsal Anomaly Definition, including:

- a) Defining anomaly scenarios to inject into rehearsals.
- b) Assist in scripting anomalies for the simulator.

6.6.5.4 Satellite Operations Exercises

The Contractor **shall** participate in Satellite Operations Exercises (SOE), including:

- a) Participate in planning of instrument-focused SOEs.
- b) Review and Approve Test Plans, Procedures, and Configurations.
- c) Support Test Readiness Review.
- d) Monitor execution of instrument-focused SOEs at Spacecraft Facility and the NOAA Satellite Operations Facility.

6.6.5.5 Compatibility Tests

The Contractor **shall** participate in satellite-level compatibility tests per instrument, including:

- a) Participate in planning of instrument-focused compatibility.
- b) Review and Approve Test Plans, Procedures, and Configurations.
- c) Support Readiness Review.
- d) Participate in Early Engineering Opportunity, dry-run, and run-for-record execution of tests.

6.6.5.6 Mission Rehearsals

The Contractor **shall** participate in Mission Rehearsals, including:

- a) Three (3) **(TBR)** instrument-focused rehearsals and dress rehearsals.
- b) Three (3) **(TBR)** Support Operations Readiness Exercises as needed to prepare for and/or supplement Mission Rehearsals.

6.6.5.7 Operations Product Validation

The Contractor **shall** support Operations Product Validation, including:

- a) Review Procedures (PROC) for consistency with ROPs.
- b) Review other ops products such as pages and tasks for accuracy and contingency procedures.
- c) Witness execution of final PROCs to assess readiness for flight.

6.6.5.8 Recommended Operating Procedures

The Contractor **shall** develop Instrument ROPs (OO-5) and support:

- a) Recommend ROP changes as needed based on lessons learned.
- b) Review any Mission Operations Team ROP changes based on operational lessons learned.
- c) Update PROCs as needed.

6.6.5.9 Operational Readiness Test

The Contractor **shall** support Operational Readiness Test (ORT) that includes the review of instrument-specific information in ORT Plan for accuracy and completeness.

6.6.5.10 Instrument Training

The Contractor **shall** provide Instrument Training Material (CDRL OO-6) and instruction. This includes nominal and contingency operations instruction to include Science Objectives, Sensor Overview, Subsystems/Layout, Day/Week/Month/Year in the Life (as appropriate), Spacecraft/Sensor Interface, Data Descriptions, Telemetry Displays, Sensor Specifications, Data Uploads/Dumps, Constraints, Considerations, Ops alerts, and Limits.

6.6.6 Commissioning and Post-Launch Support

The Contractor **shall** support Satellite Launch, Early Orbit, and Anomaly checkout, including providing necessary engineering support at the operations facility for on-orbit activation and checkout of the instrument. The Contractor **shall** provide at a minimum:

- Instrument support during bus commissioning at the operations facility (monitoring instrument survival telemetry, spacecraft out-gassing, and other instrument concerns as needed). Commissioning reports **shall** be provided in accordance with CDRL OO-3.
- Instrument participation in real-time go/no-go decision-making during instrument activation at the operations facility.
- Support during buy-off.
- Anomaly resolution support.
- Support during instrument calibration and validation.
- Verification of SMBA health, functionality, and performance. Evaluations to include trending and comparisons to requirements and expectations established during ground test and characterizations. The results of all evaluations **shall** be documented in Commissioning Report (CDRL OO-3).

6.6.7 Post Delivery and Post-Launch Sustaining Engineering and Cal/Val

The Contractor **shall** support the mission by providing:

- Anomaly resolution support through the period of performance of the Contract.
- Software maintenance through the period of performance of the Contract (includes Flight, Simulator, and GSE software).

- The Contractor **shall** support the Calibration and Validation Planning Process as an active member of the Cal/Val Planning Team.

Appendix A NPR 7150.2a TAILORED COMPLIANCE MATRIX

Compliance Matrix to be filled out by Supplier (tailored subset of NPR 7150.2a levied by NASA on the Supplier)					
NPR 7150.2a Section Title	NPR 7150.2a Req #	SWE Descriptor (See NPR SWE for full description; items with * not levied for Class C)	Compliance SMBA FSW (Class B) (Yes/No/Partial)	Compliance SMBA Sim SW (Class C) (Yes/No/Partial)	Artifact(s) Demonstrating Compliance
Software Life Cycle Planning	13	SW Plans			
	14	Execute Planning			
	15	Cost Estimation			
	16	Schedule			
	18	Reviews			
	19	Life Cycle Model			
	20	SW Classification			
	22	SW Assurance			
	23	SW Safety			
	25	Corrective Action			
	26	Plan Changes			
Off-The-Shelf Software	27	COTS, GOTS, MOTS			
Software Verification and Validation	28	Verification planning			
	29	Validation planning			
	30	Verification results			
	31	Validation results			
Government Insight	39	Insight into SW Activities			
	40	Access to SW Products			
	41	Open Source Notification			
	42	Electronic Access to Source Code			
Supplier Monitoring	43	Track Change Request			
	44	SW Measurement Data			
	45	Joint Audits			
	46	SW Schedule			
	47	Traceability Data			
Requirements Development	49	Document			
	50	SW Requirements			
	51	Flow-down & Derived Reqs.			
	52	Bi-directional Trace (parents->SW reqs)			
Requirements Management	53	Manage Requirements Changes			
	54	*Corrective Action			
	55	Requirements Validation			
Software Design	56	Document Design			
	57	SW Architecture			
	58	*Detailed Design			
	59	*Bi-directional Traceability (reqs->design)			
Legend: Gray filled items with an "*" are not levied for Class C simulator software					

NPR 7150.2a Section Title	NPR 7150.2 a Req #	SWE Descriptor (See NPR SWE for full description; items with * not levied for Class C)	Compliance SMBA FSW (Yes/No/Partial)	Compliance SMBA Sim SW (Yes/No/Partial)	Artifact(s) Demonstrating Compliance
Software Implementation	60	Design Into Code			
	61	*Coding Standards			
	62	Unit Test			
	63	Version Description			
	64	*Bi-directional Traceability (design->code)			
Software Testing	65	SW Test Plans, Procedures, Reports			
	66	Perform Testing			
	67	Verify Implementation			
	68	Evaluate Test Results			
	69	Document Defects and Track			
	70	Models, Simulations, Tools			
	71	Update Plans & Procedures			
	72	*Bi-directional Traceability (code->test)			
Software Operations, Maintenance, and Retirement	73	Platform or Hi-Fidelity Simulations			
	74	Document Maintenance Plan			
	75	Plan Ops, Maint. & Retirement			
	76	Implement Plans			
	77	Deliver Software Products			
Software Configuration Management	78	Documentation			
	79	Develop CM Plan			
	80	Track & Evaluate Changes			
	81	Identify SW Configuration Items			
	82	*Authorizing Changes			
	83	Maintain Records			
	84	*Configuration Audits			
Risk Management	85	Implement Procedures			
	86	*Continuous Risk Management			
Software Peer Reviews/Inspect ions	87	Requirements, Test Plans, Design, Code			
	88	Checklist, Criteria, & Tracking			
	89	*Basic Measurements			
Software Measurement	90	Objectives			
	91	SW Measurement Areas			
	92	Collection & Storage			
Software Documentation Contents	102	SW Development/Mgt. Plan			
	103	SW Configuration Mgt. Plan			
	104	SW Test Plan			
	105	*SW Maintenance Plan			
	106	SW Assurance Plan			

	109	SW Requirements Specification			
	110	SW Data Dictionary			
	111	SW Design Description			
	112	Interface Design Description			
	113	SW Change Request/ Problem Report			
	114	SW Test Procedures			
	115	*SW Users Manual			
	116	SW Version Description			
	117	SW Metrics Report			
	118	SW Test Report			
	119	SW Inspection/Peer Review			
Compliance	125	*Compliance Matrix			
Software Life Cycle Planning	130	*Software Safety Plan			
	134	*Safety Critical SW Requirements			
SW Implementation	135	*Static Analysis			
	136	Validation of SW Development Tools			
SW Peer Rev/Inspect.	137	Peer Review/Inspections of SW Plans			
SW Plans	138	SW Safety Plan Contents			
Legend: Gray filled items with an "*" are not levied for Class C simulator software					

Appendix B ABBREVIATIONS AND ACRONYMS

ATMS	Advanced Technology Microwave Sounder
BCR	Baseline Change Request
BOL	Beginning of Life
C&DH	Command and Data Handling
Cal/Val	Calibration/Validation
CCB	Configuration Control Board
CCP	Contamination Control Plan
CCR	Configuration Change Request
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CM	Configuration Management
COR	Contracting Officer's Representative
COTS	Commercial-off-the-Shelf
CPT	Comprehensive Performance Test
CrIS	Cross-track Infrared Sounder
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTE	Calibration Test Equipment
CTP	Consent to Proceed
CTI	Consent to Integrate
DMR	Design Modification Review
ETU	Engineering Test Unit
EGSE	Electrical Ground Support Equipment
EIDP	End Item Data Package
EMC	Electromagnetic Capability
EMI	Electromagnetic Interference
EOL	End of Life
EPR	Engineering Peer Review
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EVM(S)	Earned Value Management (System)
FM	Flight Model
FPGA	Field Programmable Gate Array
FRB	Failure Review Board
FSW	Flight Software
FSWDE	Flight Software Development Environment
FVTS	Flight Vehicle Test Suite
GEVS	GSFC Environmental Verification Specifications
GOLD	GSFC Open Learning Design
GOTS	Government-off-the-Shelf
GPR	Goddard Procedural Requirement
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
G-R	Giver-Receiver
H&S	Hardness and Survivability

Hz	Hertz
I&T	Integration and Test
IASI	Infrared Atmospheric Sounding Interferometer
IBR	Integrated Baseline Review
ICD	Interface Control Document
IMAR	Instrument Mission Assurance Requirements
IMS	Integrated Master Schedule
IMT	Integrated Mission Timeline
IRCD	Interface Requirements Control Document
IRD	Interface Requirement Document
IV&V	Independent Verification and Validation
JPSS	Joint Polar Satellite System
K	Kelvin
LEO	Low Earth Orbit
LPD	Low Earth Orbit Programs Division
LTAN	Local Time of Ascending Node
LTS	Lower-Tier Supplier
MGSE	Mechanical Ground Support Equipment
MICD	Mechanical Interface Control Document
MOR	Mission Operations Review
MOTS	Modified-off-the-Shelf
MRB	Material Review Board
MRR	Manufacturing Readiness Review
MWS	Microwave Sounder
NASA	National Aeronautics and Space Administration
NEdT	Noise Equivalent Differential Temperature
NEON	Near Earth Orbit Network
NG	Northrop Grumman
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NPSD	Noise Power Spectral Density
NS1	NEON Series-1 Project
NSOSA	NOAA Satellite Observation System Architecture
OBS	Organization Breakdown Structure
OpSim	Operational Simulator
ORR	Operational Readiness Review
ORT	Operational Readiness Test
PCB	Parts Control Board
PDR	Preliminary Design Review
PDS	Post Delivery Support
PDSP	Post Delivery Support Planning
PER	Pre-Environmental Review
PHS&T	Packaging, Handling, Storage, and Transportation
PM	Program Manager
PMB	Performance Measurement Baseline
PMP	Project Management Plan

PMR	Project Management Review
PRD	Performance Requirements Document
PROC	Procedure
PSR	Pre-Ship Review
q	Moisture
RDR	Raw Data Record
Rec-Del	Receive-Deliver
RFI	Radio Frequency Interference
RMP	Risk Management Plan
ROP	Recommended Operating Procedure
RTM	Reduced Thermal Model
SDF	Software Development Folder
SDP	Software Development Plan
SDR	System Definition Review
SEN	Software Engineering Notebook
SIIS	Spacecraft-Instrument Interface Simulator
SMA	Safety and Mission Assurance
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work
SPMM	Sensor Performance Math Model
SRF	Spectral Response Function
SRR	Systems Requirements Review
SRS	Software Requirements Specification
SSO	Sun-Synchronous Orbit
STE	System Test Equipment
SW	Software
SWAR	Software Acceptance Review
T	Temperature
TBD	To Be Determined
TBP	To Be Proposed
TBR	To Be Reviewed
TBS	To Be Supplied
TDR	Test Data Review
TIM	Technical Interchange Meeting
TPM	Technical Performance Metrics
TRR	Test Readiness Review
TVAC	Thermal Vacuum
V&V	Verification and Validation
WBS	Work Breakdown Structure